

Adverse Health Events and Vaccine Hesitancy

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Abstract

This paper studies how previous experiences with adverse health events affect vaccine hesitancy. Using nationwide Swedish administrative data linking reported adverse drug reactions to vaccination records, we first examine how individuals draw on their own past experiences when making new vaccination decisions. We analyze a severe, well-identified case: narcolepsy, a serious neurological disease plausibly induced by the 2009–2010 swine-flu vaccine. We find large reductions in COVID-19 vaccination more than a decade later, with spillovers to close family members; the effects do not attenuate among those with high health literacy or extensive prior healthcare contact. Second, we assess broader social costs of routine adverse events by studying serious events from all licensed pharmaceuticals. Here, people’s learning is narrow: adverse events from non-vaccine drugs have little effect on later vaccination, whereas serious vaccine-related events substantially reduce uptake. Together, the two analyses show that individuals rely heavily on their own experiences and that these effects are long lasting but domain-specific. Policy implications follow: while the overall impact of routine adverse events on vaccination is limited, rare but severe vaccine-related events can meaningfully lower uptake.

Keywords: Vaccine hesitancy, Adverse events, Selective memory

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1 Introduction

Recent estimates suggest that vaccines have averted up to 150 million deaths over the past half-century (Shattock et al., 2024), marking a cornerstone in public health advancements. Yet Europe has experienced an alarming resurgence in recurring outbreaks of formerly controlled communicable diseases in the past decades. The inability to contain vaccine-preventable diseases is driven mainly by insufficient demand for cheap and effective vaccination. Notably, in 2019 the World Health Organization declared vaccine hesitancy—"the reluctance or refusal to vaccinate despite the availability of vaccines"—one of the ten major threats to global health (WHO, 2019).

One potential, and arguably the most natural, source of vaccine hesitancy that prior research has largely overlooked is the perceived risk of adverse events. Understanding this channel is important because adverse drug reactions are common: Meta-analyses suggest that 5–10% of hospitalizations are due to adverse drug reactions (Insani et al., 2021; Komagamine, 2024). Beyond their immediate costs in the form of medical treatment and illness, adverse events may result in lower future uptake of vaccines and other medical treatments, imposing additional social costs—in particular through reduced uptake of vaccines against infectious diseases. Assessing the scope of these indirect social costs is essential for policies aiming to increase vaccination uptake and maintain population immunity.

Using administrative records, this paper provides evidence on the effects of exposure to adverse events on future immunization outcomes. While an existing literature links vaccine hesitancy to information gaps, trust in institutions, and peer influence, little is known about how personally experienced adverse drug events shape subsequent demand for health care later in life. Our study is the first to quantify these long-run behavioral responses using data that link individual-level adverse events to subsequent vaccination decisions.

The first part of this paper unpacks *how* individuals rely on their own experiences to inform subsequent immunization decisions. The decision to vaccinate requires individuals to trade off the expected health benefits, such as milder disease progression, against the risk of adverse events.¹ Perceptions of risks and benefits are, in turn, shaped by individuals' past experiences. The impact

¹Medical scientists and epidemiologists have long acknowledged the role of perceived costs and benefits of vaccination for the decision to vaccinate. For example, Santibanez et al. (2002) and Johnson et al. (2008) report that incorrect perceptions about the risks associated with the contraction of infectious diseases or not being immunized contribute to reluctance to vaccinate against influenza and pneumococcus.

of such experiences on future vaccination decisions depends on how individuals *generalize* across domains; an adverse event from one drug may influence uptake of that drug, related drugs, or medical treatments more broadly. Drawing on insights from cognitive psychology, when facing a decision to vaccinate, individuals use experiences that are selectively retrieved from memory. The primary driver of what experiences are retrieved and used is the perceived similarity between experiences and outcomes of decisions at hand.

To study how healthcare experiences are retrieved and shape decisions in other similar domains, we leverage cases of narcolepsy plausibly linked to a vaccine deployed during the 2009–2010 swine flu pandemic. During the swine flu pandemic, 60 % of the Swedish population was immunized with the vaccine *Pandemrix* in a national campaign. Epidemiologists later observed increased narcolepsy incidence, a chronic neurological condition with excessive daytime sleepiness and sudden muscle weakness. Subsequent studies corroborated a causal link; an estimated 150–200 individuals developed narcolepsy from Pandemrix. When COVID-19 emerged, it shared key features with the swine flu—both are fast-spreading respiratory diseases for which authorities rapidly introduced vaccines. These similarities provide a setting where previous adverse events are particularly likely to inform healthcare decisions. Drawing on these parallels, we analyze how experiencing this severe adverse event affected vaccination uptake during the COVID-19 pandemic.

In a second part of the analysis we turn to reported severe adverse events from all licensed pharmaceuticals as reported through spontaneous reporting systems² used for surveillance purposes. These reports encompass both novel and well-established treatments, making them plausibly representative of the full spectrum of routinely occurring adverse events. This exercise therefore serves to assess the overall impact on subsequent vaccine uptake from adverse events using that are broadly representative of those that patients encounter in real-world practice.

For the first part, we compare COVID-19 vaccination outcomes of individuals who developed narcolepsy after receiving the 2009–2010 swine-flu vaccine with those who received the same vaccine but did not develop narcolepsy, as well as with people who had narcolepsy for other reasons before the campaign. This difference-in-differences design isolates the effect of experiencing a severe, vaccine-induced adverse event on later vaccine hesitancy.³ For the second part, we rely on adverse events

²Most high-income countries operate such reporting systems to detect new safety signals and to gauge the frequency of adverse events after a product is introduced.

³We define vaccine hesitancy as delaying or abstaining from vaccination. Some scholars, however, use the term to describe individuals with persistent ambivalence toward vaccines—regardless of whether they eventually vaccinate

that are reported by healthcare professionals. Furthermore, the data include detailed demographic, socioeconomic, and health information for the entire Swedish population, allowing us to identify comparable individuals not only by demographics and economic background but also by prior health status as reflected in medication use and medical diagnoses. In particular, we compare vaccination outcomes of those experiencing an adverse event with observably similar individuals who received the same drug in the same year but did not experience such an event.

We find that experiencing a severe adverse event scars individuals with respect to future vaccination uptake—reducing their likelihood of getting vaccinated, the number of doses they take, and delaying the timing of the first dose. In particular, developing narcolepsy after swine-flu vaccination is associated with a 40 percentage point lower propensity to vaccinate during the COVID-19 pandemic more than ten years later compared to similar individuals. These effects spill over into personal networks of the affected. Close family members of vaccine-induced narcolepsy patients have a 10 percentage point lower propensity to vaccinate. These findings suggest that individuals are not perfectly informed but instead rely heavily on their own experiences—as opposed to officially provided information—to form an assessment about the *common* risk of experiencing adverse events from the COVID-19 vaccine. By eliciting perceived risks associated with developing COVID-19, or equivalently the benefits of preventive COVID-19 vaccination, we rule out the possibility that the results are driven by changes in the perceived benefits of vaccines rather than by heightened concerns about adverse events. In addition, individuals who have benefited from vaccinations through exposure to communicable but vaccine-preventable diseases, such as influenza (attenuated by annual vaccination), show positive effects on vaccination outcomes, highlighting that positive experiences also influence uptake. We find modest evidence of heterogeneity in the effect of exposure to narcolepsy on COVID-19 vaccination by prior healthcare experiences, suggesting that while other experiences matter, the particularly similar and salient event of narcolepsy crowds out other, likely relevant, experiences. Furthermore, we do not find smaller effects for individuals with high health literacy. While individuals with high literacy should rely more on official information (COVID-19 vaccines being safe) they are equally affected by exposure to narcolepsy. One plausible explanation is that salient experiences crowd out information, even when the information is reliable and accessible.

Experiencing a non-vaccine related adverse event in the ten years preceding the COVID-19

(see Murphy et al. (2021); Larson et al. (2014, 2022)). In their terminology, our group would instead be classified as non-compliers.

pandemic is associated with a 0.6 percentage-point reduction in COVID-19 vaccine uptake, whereas vaccine-related adverse events cause a 7.5 percentage-point decrease. These results highlight that individuals draw primarily on a narrow set of similar experiences to inform new decisions. As with narcolepsy, these effects appear long lasting; There is no evidence that only recent adverse events influence the decision to vaccinate against COVID-19. Insights relevant for policy emerge. While severe adverse events are common and their effects persist, generalization is narrow; only similar experiences have a meaningful impact on subsequent decisions. These two observations boil down to the conclusion that the overall impact of adverse events on vaccine hesitancy is limited. However, the results for vaccine-related adverse events may be relevant when the private net benefit of vaccination—ignoring potential future uptake effects—is marginal, e.g., for COVID-19 boosters.

In the final part of the paper, we ask whether the patterns we observe for newly introduced vaccines also extend to well established and safe treatments in early childhood. We find that individuals who developed narcolepsy after swine-flu vaccination are less likely to vaccinate their children with established and well-known vaccines. Similar patterns hold for adverse events from other vaccines, but not for adverse events from other types of drugs. Since childhood vaccines are well established, these findings suggest that the usual framework—where individuals rely on personal experiences when domain-specific information is scarce or hard to access—cannot fully explain the reduced uptake. Instead, we interpret these effects on immunization outcomes as suggestive evidence of a shift in the degree to which individuals rely on their own experience as opposed to external information sources; information is easily accessible but individuals do not trust the information. Finally, to assess whether scientific validation matters, we focus on autism diagnoses—a condition once erroneously associated with vaccination against measles despite the absence of any causal link. When older siblings are diagnosed with autism, parents refrain from vaccinating younger children with the MMR-vaccine. This suggests that scientific recognition is not crucial for individuals’ perception about the link between treatment and adverse event.

Our paper relates to a literature examining the effects of previous medical scandals and malpractice on future healthcare demand (Martinez-Bravo and Stegmann, 2021; Alsan and Wanamaker, 2017; Lowes and Montero, 2021; Archibong and Annan, 2023). This literature emphasizes long-term deterioration in trust following health scandals, showing large negative effects on demand for healthcare. While we also study negative experiences and healthcare demand broadly, these papers provide examples of quite extreme historical malpractice. Our data and setting allow us to assess

the marginal impact on immunization outcomes of an adverse event in a setting that is relevant for future vaccine campaigns and new pharmaceuticals. Furthermore, the administrative data allow us to study how vaccine hesitancy effects spill over in social networks, such as family members and colleagues.

Our paper is also related to a long-lasting literature on decision making under risk, and in particular, a growing literature on the lasting impact of previous experiences for decision making (see for instance Malmendier, 2021a). Our primary focus is on a *novel* risk, such as taking the COVID-19 vaccine, where individuals base their decision on a selection of related, previous experiences. A series of articles have studied how salient experiences shape the decision weights individuals assign to different events (Bordalo et al., 2012, 2022, 2024). Building on the selective-memory literature, our study contributes in three ways. First, we analyze revealed vaccination behavior rather than stated intentions. Second, we link those choices to objectively recorded experiences through comprehensive administrative data on healthcare visits and prescription-drug use rather than self-reported memories or other proxies for experiences. Third, we exploit a severe health shock whose emotional salience plausibly shaped the thinking of those affected, allowing us to quantify its long-term effect on subsequent medical decisions.

Finally, this paper is related to Oster (2018) and Giulietti et al. (2023) who focus on how exposure to vaccine-preventable diseases increases the perceived benefits and uptake of medical treatments. In relation to these articles, our primary focus is on negative experiences with healthcare rather than exposure to diseases.

The paper proceeds as follows. Section 2 reviews the swine-flu vaccination campaign and ensuing scandal. Section 3 develops a conceptual framework and testable hypotheses. Section 4 describes the data, and Section 5 explains the empirical strategy. The results unfold through three sections. In Section 7 we document effects for individuals who were affected by narcolepsy, along with heterogeneity and tests of the propositions from the conceptual framework in Section 7.2. In Section 8 we turn to the results for general severe side effects. In Section 9 and Section 10 we study effects on child vaccine immunization. Section 11 concludes.

2 Background

The Swine flu vaccination campaign The constant evolution of influenza viruses frequently leads to global transmission chains that sporadically result in severe pandemics. During the 2009–2010 A(H1N1) swine flu pandemic, several vaccines were introduced to the European market early on. The adjuvanted GSK vaccine Pandemrix became the primary, and in some countries, including Sweden and Finland, the sole vaccine used to counter the chain of infection. Amid the perceived urgency to quickly achieve high vaccination rates (Larson et al., 2022), Sweden launched a mass vaccination campaign unparalleled both in the history of the national health care system and across Europe. Survey estimates suggest that approximately 60% of the adult population was vaccinated (Holmberg and Hedberg, 2020), compared to less than 10% in countries like Germany and France that pursued more targeted strategies focusing on risk groups, such as the elderly (Mereckiene et al., 2012).

To expedite deployment, pandemrix received market authorization from the European Commission in late September 2009, after only the initial stages of clinical trials. Vaccination in Sweden began shortly thereafter, coordinated by regional health authorities. Vaccination initially targeted priority groups, such as healthcare workers and individuals at medical risk, before being extended to the general population.

In 2011, health authorities in Finland and Sweden reported a possible link between vaccination with pandemrix and an elevated risk of developing narcolepsy. Subsequent studies carried out in collaboration with Swedish authorities backed up the initial beliefs (Persson et al., 2014). This has later been corroborated with evidence from other countries. The current scientific consensus is that there is a link (Edwards et al., 2019), although developing swine flu itself may also have served as a catalyst by triggering an autoimmune response. Drawing on observed excess incidence, the Swedish Medical Products agency estimates that 150–200 people developed narcolepsy who otherwise would not have developed narcolepsy at that time (Gauffin et al., 2024).

Legal consequences and public involvement Patients who developed narcolepsy after vaccination with pandemrix were initially directed to the Swedish Pharmaceutical Insurance, a program established through an agreement between pharmaceutical companies operating in Sweden, to pursue reimbursement. As part of this program, a maximum of 150 million SEK (15 million USD) could be

paid out for all injuries attributable within one calendar year. In 2016, the government assumed the responsibility of compensating diseased individuals who had not received adequate compensation from the insurance company due to the insurance coverage limit. Individuals received at most 10 million SEK (900k USD) from the insurance company and the government combined.

The general public engaged with the vaccination campaign as it evolved into a controversy over side effects. In Panel A of Figure B2, we display the overall exposure to the topic illustrated by the number of newspaper articles in Sweden referencing the Swedish word for Narcolepsy (Narkolepsi). Panel B covers public interest as measured by the number of Google searches. Both proxies of public engagement show three major peaks in absolute numbers: (1) In 2011 around the time pandemrix was announced to be the cause of the increased incidence of narcolepsy, (2) around 2016 when the discussion on reimbursement for narcolepsy patients peaked as the government assumed the responsibility, (3) 2020 when COVID-19 cases were on the rise. Newspapers published the stories of those affected of the narcolepsy scandal, documented their endeavors to claim reimbursement, and referred back to the scandal when COVID-19 appeared on the public health agenda. The general public partook in the discussion and inquired about the disease online. These observations are meaningful to our analysis in two ways. First, there was at least minimal exposure and therefore knowledge about the narcolepsy scandal in the entire population. Naturally, this general knowledge became more salient in case of personal diagnoses or by the presence of a vaccine-related narcolepsy diagnosis in close networks. Second, it supports the idea that, in fact, the side effect of narcolepsy remained in public memory long past the time of scandal, which we implement into our conceptual framework via the role of past experiences in future health care decisions.

Narcolepsy Narcolepsy is a rare, chronic neurological disorder primarily characterized by excessive daytime sleepiness (EDS), which leads to difficulty staying awake and sudden episodes of sleep during everyday activities. There are two main subtypes: Type I narcolepsy, which includes episodes of cataplexy—sudden muscle weakness triggered by strong emotions—and Type II narcolepsy, which does not. Type I narcolepsy accounts for roughly one in five diagnosed cases. Individuals undergo extensive medical evaluations before receiving a diagnosis, and with increasingly accurate tests—such as spinal fluid analysis to measure orexin levels—misdiagnoses is expected to be rare. There is a possibility that the disease is underreported among individuals with mild symptoms, as we further discuss in section 4.

Medication can be used to manage symptoms, including EDS and cataplexy, with varying efficacy. At the same time, even with optimized medication schedule, the disease still takes a serious toll on everyday life of the diseased and requires permanent lifestyle changes. For example, in our data, among prime age individuals, those with narcolepsy have around 80% lower income than comparable individuals. Central to our analysis is the fact that the symptoms are not only salient to the affected individuals themselves, but also to those in their social environment.

During the COVID-19 pandemic, there were no official recommendations regarding the COVID-19 vaccine directed toward narcolepsy patients. We can, however, not rule out that individual practitioners may have been either advocating or discouraging the uptake of COVID vaccines. It is probable, that individuals who developed narcolepsy after pandemrix have a specific autoimmune predisposition, but this does not make them categorically more prone to experience side effects from the mRNA vaccines deployed during the COVID-19 pandemic. The mechanism by which pandemrix induced narcolepsy is suspected to have involved the immune system mistakenly attacking sleep-regulating cells in regions of the brain that produce orexin, possibly triggered by a flu protein or an adjuvant (Ahmed et al., 2014; Mahlios et al., 2013). This pathogenetic mechanism is fundamentally different from the mechanisms underlying serious adverse reactions to mRNA vaccines, such as temporary heart inflammation, which may involve a short-lived immune reaction or a rare immune misfire against heart tissue. Given this difference, the risks associated with the swine flu and COVID-19 vaccines are considered unrelated, offering no clear reason for narcolepsy patients to avoid the latter.

Narcolepsy is the only severe adverse event that was prominently discussed to be associated with pandemrix. There is no evidence to support the idea that pandemrix would be associated with more side effects than other drugs, in general. In our data, 0.002% (0.005%) report mild (severe) side effects of pandemrix. For comparison 1% (0.01%) of individuals receiving the COVID-19 vaccine reported mild (severe) side effects to the Swedish Medical Product Agency.

3 Conceptual Framework for Immunization Decisions

Healthcare settings often feature high-stakes and inevitably emotionally charged decisions in which individuals are confronted with information from a range of external sources, including authorities, personal networks, and news outlets. With technical and potentially conflicting guidance obscuring

clarity, navigating healthcare can be difficult for the layperson.

A key feature of our setting is that individuals face unprecedented decisions. The onset of the COVID-19 pandemic was characterized by great insecurity about the scope of infection and the effectiveness of vaccines. In consequence, when first deciding whether to take the COVID-19 vaccine, people were faced with limited external data on the actual risk of infection and potential vaccine side effects.

A large literature in economics examines how past experiences shape choices across settings—e.g., high-inflation episodes and stock-market shocks (Malmendier and Nagel, 2011; Malmendier, 2021a,b), and consumer demand (Bronnenberg et al., 2012). These papers document persistent effects of experience on high-stakes decisions and show that people tend to recall experiences that are emotionally salient or contextually similar to the decision at hand.

Our framework assumes that memories stem from one’s own past medical experiences (e.g., a previous flu shot) or those shared within personal networks (e.g., an anecdote about a sibling’s broken-arm treatment). Upon retrieval, different memories compete with each other; Decisions depend not only on the stock of stored experiences but on the subset that becomes accessible at decision time. We build our analysis of vaccination decisions on a simple conceptual framework of selective memory inspired by a series of papers by Bordalo et al. (2012, 2022, 2024) and that is reminiscent of the seminal paper on case-based decision theory by Gilboa and Schmeidler (1995).

The decision maker (DM) vaccinates if the perceived benefits of vaccinating $\hat{\pi}_B B$ outweigh the costs of vaccinating $\hat{\pi}_C C$. B includes both individual benefits, but may also include prosocial motives, such as protecting others from infection. We interpret $\hat{\pi}_B$ as the expected probability of not developing an infectious disease and $\hat{\pi}_C$ as the perceived probability of developing side effects from the vaccine. The perceived risk of side effects is formed according to

$$\hat{\pi}_C = (1 - \theta)\pi + \theta\hat{\pi}_E \tag{1}$$

Estimates of $\hat{\pi}$ are based on the best available public knowledge of the true probability of severe side effects π as well as $\hat{\pi}_E$, which is an assessment of π based on experiences. While we focus on the perceived risk of developing severe side effects, $\hat{\pi}_C$, we think of $\hat{\pi}_B$ as also being a combination of best available evidence and own experiences. We think of θ as the cost of acquiring information about the true risk of side effects whereas it can also be more generally thought of as the weight an individual puts on personal experiences. In particular, a lower degree of trust in healthcare

authorities would imply a higher θ . If $\theta = 0$, personal experiences such as exposure to severe adverse events, are not relevant and individuals base their decision solely on common knowledge available to them. This is the case when the true risk of vaccination is well established and the relevant information is easily accessible at low cost, such as for instance in the case of common children vaccines, which we study in section 9.

By contrast, when public signals are scarce or contradictory, acquiring reliable risk information is costly. In the initial phases of COVID-19 vaccine roll-out, little was known about the occurrence of side effects on a large scale. In the absence of perfect information on the costs of the vaccine, individuals place greater weight on personal experiences, $\hat{\pi}_E$, assessing the risks of side effects.

Following Bordalo et al. (2024), we conceptualize the decision-maker as having access to a database, E , of N experiences that shapes $\hat{\pi}_E$. We think of E as consisting of events within the healthcare domain—either personally experienced or observed through individuals in their social networks. Experiences are binary vectors of features. For example, "Hospitalized" may be one feature of an experience. Let the target event be $T = \text{"Severe side effect after COVID-19 vaccination"}$. Let, furthermore, $S(e) \in [0, 1]$ measure the similarity between an event e and the event of developing severe side effects from the COVID-19 vaccine T . Similarity increases in the number of shared features between e and T . For example, developing side effects from a COVID-19 vaccine shares many common features with developing narcolepsy from pandemrix, as both vaccines were rolled out fast during pandemics and countered versions of influenza viruses with comparable symptoms. This is in stark contrast to events of different domains, such as developing an idiopathic heart attack or even events outside the domain of diseases, such as experiencing hospitalization from injuries sustained in a car accident.

To rely on an experience, a decision maker must retrieve it from memory. We model the retrieval probability of experience $e \in E$ as

$$r(e) = \frac{S(e)}{\sum_{v \in E} S(v)} \quad (2)$$

The denominator captures interference: the more past events that resemble this target, the more they crowd each other out, reducing the chance that any single episode is retrieved.

Conditional on reactivating the memory of an event, the DM may use this event to *simulate* the event in question, such as developing side effect after vaccination with COVID-19, T . For a past experience e , let $\sigma(e) \in [0, 1]$ denote its simulation weight—how strongly e contributes to constructing the imagined outcome. Just like with recall, the ability of simulating an event is

assumed to be increasing in the similarity between two events u and v $\sigma(u) \geq \sigma(v)$ if $S(u) \geq S(v)$. That is, memories of events that are similar to the COVID-19 vaccine, such as pandemrix, are predominantly used to simulate the expected outcome of COVID-19 vaccination and thereby inform individual vaccination outcomes. In this way, σ provides a micro-foundation for how individuals use previous experiences to assess risks; σ reflects the idea that more similar experiences are easier to simulate and are given greater weight and, consequently, exert more influence on the vaccination decision. Consequently, σ also captures the idea that experiences that seem more relevant—precisely because they are easier to simulate—are preferentially used; e.g., an individual who experiences vaccine-induced side effects, rather than other side effects, will perceive vaccines as riskier than other drugs.

The experience-based estimate $\hat{\pi}_E$ is defined as

$$\hat{\pi}_E = \sum_{e \in E} r(e) \sigma(e) \quad (3)$$

The perceived risk is the average of each past experience’s simulated risk $\sigma(e)$ weighted by how likely that memory is to be retrieved $r(e)$.

The conceptual framework, where $\hat{\pi}$ depends on the database E , the cost of acquiring information θ , and officially provided information π leads to the following testable predictions, which we return to in Section 7.

Prediction 1 *Individuals with lower health literacy have higher θ and are more dependent on their own experiences.*

If θ can be interpreted as the cost or difficulty of acquiring information about π , we expect individuals with low health literacy have higher θ and hence to adjust their behavior more to adding a new experience to their database. We rely on three proxies for health literacy to test whether individuals rely on experiences to different extent: (i) having a doctor in the family; (ii) paternal cognitive ability; and (iii) years of schooling. Suppose adding e and e' both increases $\hat{\pi}_E$. Then adding e increases $\hat{\pi}_E$ more than adding e' if and only if $S(e) > S(e')$

That is, the degree to which $\hat{\pi}_E$ increases—for sufficiently similar experiences—is increasing in similarity. Due to the parallels of the swine flu pandemic and the COVID-19 pandemic, experiences from pandemrix are more likely to be recalled and used for simulation of COVID-19 vaccine outcomes than other adverse health care experiences. In particular, we test this proposition by studying

experiences that will differ in similarity but not necessarily in severity.

In addition, the conceptual framework predicts that additional memories cause interference because they enlarge the decision maker’s memory database with experiences that compete for retrieval:

$\hat{\pi}_E$ is less sensitive to adding e when E is larger.

Intuitively, a larger number of experiences crowd-out the probability of recall for a particular experience when assessing the probability of T . The data allow us to directly measure the number of other experiences with the healthcare system. In particular, we test Section 3 by considering heterogeneity in the effects on immunization outcomes of exposure to side effects with respect to number of other experiences as measured by number of healthcare visits and number of drugs taken.

Research in cognitive psychology suggests that other internal factors, such as emotions and identity, connected to memorized events also play a crucial role in the forecasting of decision outcomes. First, emotional events are more likely to be retrieved from memory than neutral events (LaBar and Cabeza, 2006). Second, emotions are encoded with the event and, when the memory is recalled, that affective ‘tag’ is reinstated and amplifies the experience’s impact on the imagined side-effect outcome (For a recent example in economics integrating the role of emotions in memory, see (Ashraf et al., 2024)). While we refrain from modeling emotions explicitly in our conceptual framework, the severity of the condition and the uncertainty that revolved reimbursement from governmental agents the scandal likely sparked a range of negative emotions. Figure B2 supports the idea of strong emotional tagging. Local newspapers wrote personal and in-depth stories about the affected individuals, often young children along with their families, were affected by the disease and their struggles fighting both mistrust among the general public and the government. Because personal relationships increase the emotional exposure to severe medical conditions, such as narcolepsy, we hypothesize that immediate networks of the diagnosed patients carry stronger effects than more distanced members of personal networks. In particular, we would expect direct family members, who first-hand experience the restrictive nature of the disease, to be most affected by the emotional burden of a vaccine related narcolepsy diagnosis.

4 Data

4.1 Administrative Records

The data used for the analysis in this paper is accessed through the Swedish Register-based Research Program on COVID-19 (SWECOV). Permission to use the data is obtained from Sweden’s Ethical Review Authority (permit numbers 2021-02225, 2022-013550-02, 2022-06118-02 and 2024-02342-02).

We combine data from several Swedish administrative sources covering the Swedish population. First, we use data on drugs and healthcare visits from the Patient register and the Drugs register, administrated by the National Board of Health and Welfare. These sources encompass information on all specialist care visits, diagnoses, and drug prescriptions, covering the period 2005–2022. To identify individuals with narcolepsy we use detailed diagnosis and drug codes on narcolepsy diagnoses and associated pharmaceutical treatments during this period. In particular, we have access to the full diagnosis codes (ICD-10-SE) for narcolepsy, G47.4A–G47.4X and drugs commonly taken by individuals with narcolepsy. The remaining drug and disease codes are truncated. This allows us to pin down individuals with narcolepsy, regardless of whether they developed it from pandemrix or not.

The Public Health Agency provides us with data on all COVID-19 vaccinations in Sweden up until March 2023. The data entails information on the date of each administered vaccine dose as well as the brand and manufacturer of the vaccine, which allows us to construct our main outcome variables of COVID-19 vaccine hesitancy. Furthermore, The Public Health Agency provides us with novel data from the Swedish Children’s vaccination program between 2013 and 2024. We use data on measles, pneumococcus, and diphtheria vaccinations administered during the first two years of life⁴. We also use phone calls to 1177, a medical advice helpline run by the Swedish regional healthcare authorities, from which we observe phone calls related to COVID-19 between 2019 and 2023. We access information on COVID-19 tests that were administered by the public healthcare system during the initial period of the pandemic. We use the data on tests and phone calls to elicit perceived risks associated with developing COVID-19.

The Swedish Medical Products Agency supplies records of every suspected adverse drug reaction

⁴The measles vaccine is almost always given in combination with the mumps vaccine and the rubella vaccine, combined called the MMR-vaccine. Throughout this paper we’ll focus on measles vaccinations, but it will typically coincide with vaccination against also mumps and rubella.

reported through the national spontaneous-reporting system between 2005 and 2024. Each report includes date of onset and date of notification, reporter category (health-care professional or lay person), the suspected medicinal product (ATC code and brand name), MedDRA-coded reaction terms, and the regulatory seriousness classification used in EudraVigilance. Although under-reporting is substantial, all physiological reactions to medication that are secondary to its therapeutic purpose are expected to be reported to the system.

Information on pandemrix vaccinations are accessed through registers maintained by the regional health authorities. To build our focal sample of individuals who received pandemrix, we gather vaccination data from 10 of the 21 regions (see Table D4 for a breakdown of the different regions along with vaccination rates). Out of the remaining 11 regions, nine regions did no longer have access to the data in an accessible format and two regions were not willing to provide the data. In total, the regional pandemrix dataset covers 13% of Sweden’s in 2010.

We use register data administrated by Statistics Sweden on socioeconomic and demographic characteristics, such as occupation, income, family links, and place of residence. We use this data to define individual networks that are used to analyze how vaccine hesitancy spreads beyond the diseased individuals. In addition, it allows us to identify individuals comparable to those who developed narcolepsy or who were exposed to someone that developed the disease using matching techniques. Finally, we access data from the Swedish Military Archives on scores from draft screening tests that individuals conducted between 1979 and 1997, a period when military enlistment was mandatory for men. We use this data to derive a measure of *Cognitive Ability*.

4.2 Main Samples & Key Variables

We define individuals in our treatment group as as having pandemrix-induced narcolepsy if they (i) reported narcolepsy as a side effect and (ii) are diagnosed with narcolepsy in specialized care at least once after November 1st 2009 but not before that. We consider all individuals who received the diagnosis G47.4 (ICD-10-SE) and hence do not make any restrictions on the type of narcolepsy developed. This yields a sample of 346 individuals. Panel A of Figure 1 displays the distribution of age at the time of first narcolepsy diagnosis. Individuals were on average 17 years old when they were first diagnosed with narcolepsy. Most of them report narcolepsy symptoms within three years past vaccination. Based on excess prevalence calculations, about half of the individuals in this sample would have developed narcolepsy even if they abstained from vaccination with pandemrix.

The pathogenesis of narcolepsy is rather complex and it is impossible to isolate the individual cause of the disease; It is likely that all patients in the treatment group attribute their disease in parts to pandemrix. We observe a slight increase in narcolepsy diagnoses following the swine flu pandemic also among individuals who do not report it as a side effect. These are likely individuals who had some mild form of narcolepsy before the pandemic but did not associate their particular symptoms with a specific health condition. In addition, narcolepsy is a rare condition, and even general practitioners may not have been familiar with its specific symptomatology before narcolepsy became salient in the media.

[Figure 1 about here]

We define a "pre-swine flu" sample of individuals serving as a placebo control group, which are diagnosed with narcolepsy between January 1st 2005 and October 1st 2009—just before the introduction of the vaccination with pandemrix. This placebo control group consists of 847 individuals who developed narcolepsy prior to the swine flu pandemic and could, therefore, not have developed it from pandemrix. Typically, a diagnosis is recorded when an individual visits the healthcare system for symptoms related to the disease at hand, which means that this sample also includes individuals who developed narcolepsy much earlier than 2005.⁵ These individuals are on average 37 years older than those with pandemrix-induced narcolepsy. We impose the additional restriction that these individuals did not report any other side effects from pandemrix.

We contrast medication take-up of narcolepsy patients in the treatment group and the placebo control group to better understand potential differences in clinical manifestations between the two groups. Figure B3 shows the development of prescriptions for common medications among the two groups. In line with observations in Gauffin et al. (2022), patients in the post-swine flu sample are in large parts immediately prescribed modafinil, methylphenidate and sodium oxybate. This combination represents the to-date most common pharmaceutical routine for narcolepsy patients. Furthermore, also in line with Gauffin et al. (2022), the prescriptions of dexamphetamine and lisdexamphetamine increased after 2014. Accordingly, the prescribed drug schedules are slightly different in terms of the aforementioned medications for the post-swine flu and the pre-swine flu sample. Changing medication routines takes time, and patients often get started on new prescriptions while fading out the existing medical treatment. For instance, Sodium oxybate, an anesthetic treating cataplexies, did not become available on the market until 2012, implying that

⁵It is estimated that around 4 000 individuals in Sweden suffer from some type of narcolepsy (Gauffin et al., 2022).

patients with narcolepsy-induced pandemrix were prescribed with it right away while as patients diagnosed before 2010 either stuck to established routines or switched medication later. The differences is also partly explained by the age difference between the two groups. For example, sodium oxybate and atomoxetine are typically not prescribed to older individuals due to elevated risk of respiratory depression. In sum, we do not find any clear signs that either of the two groups would have more severe narcolepsy symptoms based on prescriptions.

Measures of vaccine hesitancy We consider vaccine hesitancy along three different margins (i) whether an individual took at least one COVID-19 vaccine dose, (ii) the number of doses conditional on taking at least one dose, and (iii) the time elapsed between the dose becoming available to an individual and timing of the first dose. We restrict all analyses to individuals living in Sweden in 2021. Accordingly, we define an individual as unvaccinated if they lived in Sweden in 2021 and do not have a registered COVID-19 vaccination dose. We take on a data-driven approach to define a date when the vaccine is first available to an individual. We define a date of availability for each combination of birth year and region. For each birth-year $\times$ region cell, let $f(i)$ denote the vaccination-date of the i -th individual to be vaccinated (Excluding healthcare workers). We define the date of availability as $f(i^*)$ where $i^* = \arg \min_{i \leq N-50} \{f(i+50) - f(i)\}$. In words, the date of availability is the date that minimizes the number of days elapsed between individual i 's vaccination and the vaccination date for individual $i+50$. In ?? we display the distribution of first vaccinations across time along with defined first date of availability for a sample of 20 region $\times$ birth year-cells. In general, the distributions of first vaccinations within these region $\times$ birth year-cells are unimodal and concentrated.

Personal networks We differentiate between close and extended networks. Close networks consist of direct *family*, including biological parents and full siblings of the focal individuals, as well as *extended family*, including biological cousins, uncles, aunts and grandparents of the diseased individual. We consider three extended networks: *neighbors* defined as individuals that lived in the same DeSO region as the sample individuals in 2011⁶ *schoolmates* of the diseased individual in 2011; *colleagues*, that work at the same plant/establishment in 2021 as a sample individual and who are not family members of the treated/control individuals. If an individual belongs to at least

⁶A DeSO region is the smallest geographical unit that we have access to and consists of x number of individuals on average for our sample.

one network of a treated individual, the network member is classified as treated. We assign each network member to only one individual in the focal sample.

Classification of Adverse Events Throughout this paper, we classify side effects into three mutually exclusive categories. In particular, we distinguish between two types of vaccine related side effects, namely pandemrix-induced narcolepsy as well as other side effects from vaccines. In addition, we consider the entirety of non-vaccine induced side effects which we refer to as general side effects. We restrict the attention to severe side effects. A side effect is classified as severe if the individual sought healthcare for the experienced symptoms. We include side effects that are classified as self-reported, as well as those reported by medical practitioners. We describe the most common reported general severe side effects in Table D3. 80% of general severe side effects are reported by healthcare professionals and the remaining 20% are self-reported. Evidently, the most common adverse general side effects are transient conditions without serious medical consequences.

5 Empirical Design

Our setting comes with two main empirical challenges. The first challenge—which is primarily relevant to the analysis of individuals with narcolepsy—is to isolate the effects from drug-induced side effects to the effects of having a condition that was not attributed to a drug; We think of *having* a condition such as narcolepsy as a mediating factor of the vaccine hesitancy effect from exposure to adverse events. For example, individuals with narcolepsy may have different experiences with the healthcare system or different social networks that shapes their attitude toward the COVID-19 vaccines. The second challenge, which is relevant for the analysis of exposure to adverse events in general, relates to the fact that we observe reported adverse events rather than actual adverse events. In Section 5.2 we discuss under what assumptions the estimated effect may be interpreted as the average effect of developing side effects.

5.1 Isolating hesitancy-effect

We are interested in the effects on immunization outcomes of developing narcolepsy that may have been caused by pandemrix. A natural estimator for this effect is

$$\tau_{ideal} = \mathbb{E}[Y_{ith}|t = \text{Pandemrix}, h = \text{Narc. after}] - \mathbb{E}[Y_{ith}|t = \text{Pandemrix}, h = \text{Narc before}]$$

In words, we would ideally compare individuals who develop narcolepsy from pandemrix to individuals who took pandemrix and developed narcolepsy prior to pandemrix vaccination, and who hence could not have developed it from pandemrix. Because narcolepsy is rare, the overlap between individuals diagnosed with narcolepsy and those observed receiving Pandemrix is small; the data contain too few cases of individuals who both develop narcolepsy and subsequently receive Pandemrix. Instead, we construct a difference-in-difference estimator that allows us to rely on all narcolepsy cases prior to the swine-flu pandemic. Expanding the previous expression yields

$$\begin{aligned} \tau_{ideal} = & (\mathbb{E}[Y_{ith}|t = \text{Pandemrix}, h = \text{Narc. after}] - \mathbb{E}[Y_{ith}|t = \text{Pandemrix}, h = \text{No narc.}]) \\ & - (\mathbb{E}[Y_{ith}|t = \text{Pandemrix}, h = \text{Narc. before}] - \mathbb{E}[Y_{ith}|t = \text{Pandemrix}, h = \text{No narc.}]) \end{aligned}$$

The first difference is the difference between individuals who develop narcolepsy and individuals who take pandemrix and do not develop narcolepsy. The second difference is the difference between individuals who develop narcolepsy *before* the swine flu pandemic and individuals who take pandemrix and do not develop narcolepsy. Once again, the issue is that we can not credibly estimate $\mathbb{E}[Y_{ith}|t = \text{Pandemrix}, h = \text{Narc. before}]$ because we observe too few individuals with narcolepsy who also receive pandemrix. Instead, the key assumption is the following

Assumption 1 *Constant narcolepsy effect*

$$\begin{aligned} & \mathbb{E}[Y_{iht}|t = \text{Pandemrix}, h = \text{Narc. before}] - \mathbb{E}[Y_{iht}|t = \text{Pandemrix}, h = \text{No narc.}] \\ = & \mathbb{E}[Y_{ith}|h = \text{Narc. before}] - \mathbb{E}[Y_{ith}|h = \text{No narc.}] \end{aligned}$$

That is, the effect on vaccinations of developing narcolepsy is independent of pandemrix-status. Put differently, the effect of developing narcolepsy before the swine flu pandemic is the same as among pandemrix-takers.

This assumption allows us to estimate the following object

$$\begin{aligned} \tau_{dd} = & \overbrace{(\mathbb{E}[Y_{iht}|h = \text{Narc. after}, t = \text{Pandemrix}] - \mathbb{E}[Y_{iht}|h = \text{No narc.}, t = \text{Pandemrix}])}^{\tau_{post}} \\ & - \underbrace{(\mathbb{E}[Y_{ith}|h = \text{Narc. before}] - \mathbb{E}[Y_{ith}|h = \text{No narc.}])}_{\tau_{pre}} \end{aligned} \tag{4}$$

Intuitively, we compare vaccinations of individuals who take pandemrix and develop narcolepsy to individuals who take pandemrix, filtering out any potential effects of having narcolepsy that is not pandemrix-induced.

τ_{dd} may be interpreted the causal effect of developing pandemrix induced narcolepsy under a conditional independence assumption:

Assumption 2 *Conditional independence*

$$\mathbb{E}[Y_i(0)|h_i = \text{Narc.}, X_i] = \mathbb{E}[Y_i(0)|h_i = \text{No Narc.}, X_i] \quad (5)$$

In words, conditional on a set x_i of observable characteristics, developing a side effect is as good as random. We return to the plausibility of this assumption in Section 5.3

5.2 Selection into Reporting

We wish to estimate the causal effect on vaccination of developing a side effect such as narcolepsy. We, however, estimate the effect of developing *and* reporting side effects. This parameter is difficult to interpret as the econometrician can not manipulate reporting.

Consider an individual who may develop a drug side-effect. Let $W_i = 1$ if an individual develops a side-effect and 0 otherwise. In general W_i is unobserved to both the individual and the econometrician. The individual, however, forms a subjective belief about the probability that she had a side effect based on cues she receives such as experiencing symptoms, information provided by the doctor or through media.

$$p_i = \Pr(\text{“drug caused symptom”} \mid W_i = 1, X_i),$$

And reports the symptom as a side effect if she think it was likely enough to have been a side effect:

$$D_i = \begin{cases} 1, & p_i > \bar{p}, \\ 0, & p_i \leq \bar{p}, \end{cases}$$

where $\bar{p} \in [0, 1]$ is an arbitrary constant. Similarly, the agent may have some symptom that she wrongly attributes as a side effect:

$$q_i = \Pr(\text{“drug caused symptom”} \mid W_i = 0, X_i), \quad D_i = 1 \iff q_i > \bar{p}.$$

Let $Y_i(w)$ be the outcome under true state $W_i = w$. We estimate the average difference among reporters and non-reporters:

$$\Delta_D(X) = E[Y \mid D = 1, X] - E[Y \mid D = 0, X]$$

While as the object of interest is the average difference between those that develop a side effects and those that do not:

$$\Delta_W(X) = E[Y(1) \mid W = 1, X] - E[Y(0) \mid W = 0, X].$$

The goal is to connect $\Delta_W(X)$ to $\Delta_D(X)$. We invoke the following assumption:

Assumption 3 *Equivalence of reported and unreported side-effect cases*

$$E[Y(0) \mid W = 0, D = 1, X] = E[Y(1) \mid W = 1, D = 1, X].$$

Among reporters, the effect among developers and non-developers is the same in expectations. This is plausible since individuals, conditional on observable characteristics, do not know if they developed a condition as a side effect or would have developed it anyways. Put differently, conditional on receiving the cue that a symptom was a side-effect, actually having it makes no difference. Furthermore, we make use of the following assumptions

Assumption 4 *Missing-at-Random among treated and untreated (MAR)*

$$p_i \perp Y(1) \mid W = 1, X, \quad q_i \perp Y(0) \mid W = 0, X.$$

In words, MAR among treated states that (conditional on X), among those that develop side effects, reporters and non-reporters would have behaved the same in the absence of reporting. Put differently, conditional on developing a side effect, the perceived probability p_i that the condition was a side-effect is independent of future vaccine hesitancy. Similarly, MAR among untreated states that, among individuals that do not develop a side effect, reporting is independent of vaccine hesitancy.

Assumption 5 *rare side effects* $\Pr(W = 1 \mid X) \ll \Pr(D = 0 \mid X)$

The number of individuals developing side effects is small relative to the number of non-reporters. reported side effect-equivalence together with $\text{mar}(1)$ directly give us

$$E[Y(1) \mid D = 1, X] = E[Y(1) \mid W = 1, D = 1, X] = E[Y(1) \mid W = 1, X]$$

Rare side effects together with $\text{mar}(2)$ give us

$$E[Y \mid D = 0, X] \approx E[Y \mid D = 0, W = 0, X] = E[Y \mid W = 0, X]$$

such that

$$\Delta_D(X) \approx \Delta_W(X)$$

In summary, we need four ingredients. First, among individuals who truly experience the side effect, the decision to report it must be ignorable once we condition on observed covariates; some receive salient cues (such as developing narcolepsy) that link their symptoms to the drug, whereas others do not. In particular, among those who develop the side effect, the decision to report it is unrelated to unobserved traits like vaccine hesitancy. Second, the same ignorability must hold for those who did not develop the side effect: conditional on observables, whether they mistakenly attribute their symptoms to the drug is likewise random. Third, conditional on reporting, the subsequent outcome effect is assumed to be the same for true developers and for mistaken reporters—intuition being that once a symptom is viewed as drug-related, the underlying biological status no longer differentially influences future behavior or health. For instance, a person who would have developed narcolepsy regardless of vaccination reacts no differently than someone whose narcolepsy was vaccine-induced, because neither can disentangle the true source of the condition. Finally, the pool of non-reporters consists almost entirely of genuine non-developers, with only a negligible share of false negatives. Taken together, these assumptions ensure that the average difference in outcomes between reporters and non-reporters recovers the causal effect of actually developing the side effect.

Individuals who developed narcolepsy before the swine-flu pandemic may display a different propensity to take the COVID-19 vaccine for two distinct reasons. First, they may believe they are more susceptible to side effects and, by affecting educational and occupational trajectories, alter the social contexts that in turn shape vaccination behavior. Second, these individuals may be more hesitant because they are more aware of the health scandal. We refer to this latter channel as an *information effect* which, obviously, is an outcome of the health scandal. It is challenging to discriminate between the information effect and the narcolepsy effect⁷.

We can make progress on which mechanism is more plausible by considering other severe, chronic and neurological diseases. Figure 2 displays differences in vaccination rates between individuals with a set of other diseases, controlling for age and gender, along with the two estimates for τ_{post} and τ_{pre} .

[Figure 2 about here]

Individuals that developed narcolepsy before the swine flu pandemic have a lower COVID-19

⁷More generally, this violation of SUTVA also applies to individuals without narcolepsy.

vaccination rate than individuals with other similarly severe diseases. We can not rule out that there may be something specific about having narcolepsy as compared to other diseases that prohibits them from receiving their vaccine shot. For example, due to restrictive daily schedules from day-time-sleepiness or exterior influences, such as recommendations from medical personnel. The results in Figure 2, however, support the idea that individuals with narcolepsy before the swine flu pandemic are primarily affected by being more aware of the scandal and the risk of severe side effects rather than thinking that narcolepsy serves as a contraindication for taking the COVID-19 vaccine.

5.3 Balance and matching

As is evident from the previous subsection, our empirical approach requires fairly strong independence assumptions about individuals with pandemrix-induced narcolepsy. The primary concern is that affected individuals are selected in terms of (i) developing and being diagnosed with narcolepsy, and (ii) reporting narcolepsy as a side effect. We discuss these concerns in turn.

Characteristics of individuals with narcolepsy diagnoses First, although the pathogenic mechanisms that trigger narcolepsy are still not well understood, the medical literature has found no comorbidities that would systematically predispose particular sub-populations to the disorder. Differential incidence driven by underlying biology therefore appears unlikely. Second, the sharp rise in public awareness after the pandemrix controversy, together with the fact that diseased individuals require medical treatment, makes it improbable that only a selective subset of patients sought care and received a diagnosis. We quantify how individuals with narcolepsy differ from the general population using extreme gradient boosting (XGboost), a machine-learning algorithm that sequentially adds decision trees. In particular, we fit a model predicting first-time narcolepsy diagnoses after 2016 using socioeconomic characteristics, and pre-diagnosis health history. The time period set five years after swine flu pandemic reduces the chance that cases stem from pandemrix and instead isolates factors linked to developing and being diagnosed with narcolepsy; it also lets us observe health conditions before onset. Model performance is summarized by the area under the receiver-operating-characteristic curve (AUC), which ranges from 0.5 (no predictive power) to 1 (perfect prediction). The model is estimated on a subsample that is balanced in terms of birth year.⁸ The AUC is therefore interpreted as a measure of the ability to predict who is treated, above

⁸Specifically, each treated unit is matched to 100 untreated units with the same birth year.

and beyond birth year.

As shown in Table B1, this exercise yields an out-of-sample AUC of about 0.55, implying limited correlation between developing narcolepsy and any set of observable socioeconomic and health characteristics. We benchmark these numbers against an unrelated health condition, namely the incidence of a heart attack (I20, I21, I50, ICD-10-SE), which is known to be associated with socioeconomic characteristics as well as individual morbidity profiles (Adhikary et al., 2022). Fitting a model including personal socioeconomic and health history on 365,000 individuals who developed a heart attack after 2015 along a random sample of individuals residing in Sweden in 2021 information yields a relatively higher AUC of 0.61. Taken together, neither health characteristics or socioeconomic characteristics

[Table B1 about here]

Non-random reporting of side effects Another challenge is potential selection in reporting narcolepsy as a side effect (ii). Importantly, patients had financial incentives to reporting narcolepsy as a side effect to meet the criteria for the reception of government and insurance reimbursement.

Yet, the remaining main concern is that there may be latent vaccine hesitancy, orthogonal to observable socioeconomic and health characteristics, that is correlated with being diagnosed with and reporting narcolepsy as well as behavior during the COVID-19 pandemic.

We proceed by quantifying how different treated individuals are in terms of observable characteristics compared to the control group. In ?? we display differences in the covariates between the diseased individuals and the control individuals, conditioning only on birth year. Individuals in the treatment group have higher income and years of schooling. There is no clear pattern regarding their health status, on the one hand they have slightly fewer drug prescriptions and less sick leave but they also make more healthcare visits. Diseased individuals live in more urban areas, which is likely an artifact of the fact that we do observe control individuals from the larger metropolitan areas. Individuals born before or in 1990 (panel b) report narcolepsy as a side effect somewhat later than individuals born after 1990 (panel a) but appear to not be more selected in terms of socioeconomic characteristics, despite the fact that these are individuals who were more likely to have developed narcolepsy anyways.

In ?? we display the same coefficients for the pre-sample. Recall that, for the pre-sample, we compare individuals who were diagnosed with narcolepsy prior to the swine flu pandemic to

individuals of the general Swedish population, independent of pandemrix status. The differences between these individuals and the general population are similar to the differences in ???. For example, individuals who developed narcolepsy before the swine flu pandemic have parents with more years of schooling than the general population.

Matching We use matching techniques to identify individuals that are comparable to those with Pandemrix-induced narcolepsy (treatment group) and to those that developed narcolepsy prior to the swine flu pandemic (placebo control group). In our preferred method we use 1:1 propensity score matching without replacement, exact matching on year of birth and gender, and propensity scores computed using logistic regression. To assure the robustness of our approach, we consider two alternative matching procedures, namely computing propensity scores with gradient boosting and coarsened exact matching (CEM) instead of propensity score matching. The former aims at better handling interactions, accommodating functional forms that are not well captured by the logit model. The latter aims at dealing with the fact that we have few treated units, making the maximum-likelihood estimates in the logit propensity-score model unstable. Instead, matching is done exactly on coarsened versions of the continuous covariates.

[Figure ?? about here]

In Figure B6 we display how the different methods perform in terms of achieving balance across the covariates. The differences in performance are small. If anything, CEM performs somewhat worse than the NN and gradient boosting propensity score matching in achieving balance.

For the main sample, we use a set of candidate matching variables consisting of characteristics of individuals and their parents as measured between 2005 and 2009, right before individuals may have developed narcolepsy symptoms due to Pandemrix. For individuals that are born after 1990 we restrict the matching variables to time-invariant and parental characteristics as we cannot measure socioeconomic characteristics for the individuals themselves between 2005 and 2009, as they are too young to observe income, years of schooling, etc. Out of the treated individuals, 278 are born 1990 or later, 77 are born before 1990. We also use parental and time-invariant characteristics for the pre-swine flu sample, as socioeconomic characteristics are likely influenced by the presence of narcolepsy (recall that we do not observe when these individuals first developed narcolepsy). The set of candidate covariates is listed and defined in Appendix D. We use a simple LASSO algorithm with cross-validation to select variables to match on. We use this procedure separately for individuals born

1990 or later and those born before 1990, for the pre- and post-samples as well as for the different networks, implying that the variables used for matching varies across these dimensions. Matching is always done on the characteristics of the network members rather than the characteristics of the focal individuals. Note, that although we ensure that the focal individuals have received pandemrix, we make no such restrictions on the network members.

6 Quantifying the Aggregate Role of Healthcare Experiences

The conceptual framework is motivated by the idea that personal experiences influence vaccination decisions, a notion we explore by providing descriptive evidence on the role of prior experiences during the COVID-19 pandemic. In Figure 6 we display AUC values for predicting whether or not individuals take at least one COVID-19 vaccine dose. We restrict the sample to individuals between age 40 and 60 by 2021 and who, based on their diagnosis history, were not in a COVID-19 risk group. This restriction implies that there is little medical reason for the individuals to not take the COVID-19 vaccine. We predict their COVID-19 vaccination status based on all health related variables that we observe, namely their diagnoses, medical drugs and reported side effects between 2010 and 2020. Our preferred method to compute propensity scores is a recurrent neural network. The recurrent network explicitly takes into account the ordering, although not the timing, of events. We also use xgboost but consistently get somewhat smaller AUC values. We benchmark the prediction score against what we get using (time-invariant) socioeconomic characteristics. The estimation sample contains 2.84 million individuals (90% train / 10% test).

[Figure 6 about here]

We find that health history alone yields an AUC of 0.65, Socioeconomic variables yields 0.72 and the full model, including both health and socioeconomic variables, yields a value of 0.76. These numbers are similar to what is found in a recent study using Finnish data Hartonen et al. (2023) who find an AUC of 0.8, using data similar to ours but where the authors also access data on genes and include vaccination status of family members. Similar to us, they find that, apart from income, medication history has the highest predictive value. The prediction exercise provides descriptive evidence that previous experiences with healthcare at large shape attitudes toward vaccines. Taken together, while socioeconomic characteristics strongly predicts vaccination uptake, the incremental increase from a decade of personal health experiences is substantive, lending empirical support to

the importance of personal experiences.

7 Exposure to narcolepsy

7.1 Main Results

Effects among diseased individuals and family members We document the effects on our main COVID-19 vaccination outcomes in Figure 7. These include: (i) whether an individual takes the COVID-19 vaccine, (ii) the number of doses conditional on taking one dose, and (iii) the time until the first vaccine dose. We present results both for individuals who themselves developed narcolepsy and for their family members. We report results for τ_{post} , the mean difference between treated individuals (post–swine flu narcolepsy) and matched individuals that received pandemrix; τ_{pre} , the mean difference between placebo control individuals (pre–swine flu narcolepsy) and matched individuals from the general Swedish population; and τ_{dd} , the difference between τ_{post} and τ_{pre} .

Individuals who developed narcolepsy were 40 percentage points less likely to have received at least one vaccine dose than their matched controls, among whom 92 percent were vaccinated. We also find a lower vaccination rate for the placebo control group of individuals that developed narcolepsy before vaccination with pandemrix. As discussed in Section 5, it is unlikely that there is anything inherent about individuals with narcolepsy. Instead, the plausible explanation is that these individuals are more aware of the medical scandal and therefore more hesitant. Combining τ_{post} and τ_{pre} yields a large effect of 35 percentage points lower vaccination rate. Given the informational spillovers to individuals that developed narcolepsy before the swine flue pandemic, we interpret this as a lower bound of the effect of developing pandemrix-induced narcolepsy.

To gauge the economic significance, we compare the magnitude of the effects to the socioeconomic gradient in vaccination rate which is displayed in Figure B1. There is a meaningful gradient in immunization outcomes along socioeconomic characteristics but it is small compared to the effect sizes that we find. For example, the difference in vaccination rate between individuals with post-tertiary schooling and individuals with only primary school education is no higher than five percentage points. These differences are large given the high baseline vaccination rate, but small relative to the effect sizes of being exposed to severe side effects.

We next consider the number of doses, conditional on getting one dose, to account for vaccine

hesitancy expressed in incomplete vaccination schedules (intensive margin). We find differences of 0.1–0.4 (2.8 doses in the matched control group of diseased individuals) fewer doses. These coefficients are small in magnitude compared to the effects on the extensive margin. In other words, individuals willing to take the first dose seem to be less hesitant to also take additional doses required to meet the full vaccine schedule.

Considering the number of days elapsed until the first vaccination, diseased individuals delay their first vaccination by approximately 40 days, family members by approximately a week, and extended family members by about three days. The fact that affected individuals delay their vaccination is an indicator that they are concerned about side effects and prefer to wait to see if others experience side effects from the COVID-19 vaccine. Once again, there is a socioeconomic gradient in time until first dose; individuals with only primary education (Nine years of schooling) get the vaccine 10 days later than individuals with post tertiary education. This is similar to the magnitude of the effect for family members.

We replicate the results in Figure 7 using coarsened exact matching and propensity scores computed using gradient boosting in ?? and Figure E1. Our results remain virtually identical with the exception of *Number of Doses* where we find negligible effects for CEM. In Figure B8 we address the concern that placebo control individuals are significantly older than treated individuals. This is a concern if the mediating effect of *having* narcolepsy differs by age. We display one estimate where we reweigh the birth year distribution of narcolepsy individuals in the placebo control sample to match the birth year distribution of the treated units in the main sample. To handle the fact that the overlap in age is small between the two groups we next perform an exercise where we extrapolate the effect from the age distribution of treated units in the placebo control sample. None of these two exercises changes the estimate of τ_{pre} meaningfully. This suggests that our estimated effect of having narcolepsy in the placebo control group reflects the counterfactual effect of developing narcolepsy, that was not pandemrix-induced, at the age of when individuals developed pandemrix-induced narcolepsy.

[Figure 7 about here]

In light of our framework, the difference in magnitude of the results between direct family members and extended family members is puzzling. Both direct and extended family members are regularly exposed to the individual with narcolepsy, and will know that it may have been induced by pandemrix. Direct family members, however, are more likely to witness the patient’s sudden change

in lifestyle, regular symptoms and daily challenges, which makes the condition much more salient for them. This is particularly true, because direct family members bear the caregiving responsibilities of the diseased, such that their memories of these struggles become more vivid and influential in shaping subsequent health decisions. We further hypothesize that the heightened emotional tagging in closely related family of the patients makes the experience particularly easy for them to recall and use for simulation.

In Table C1 we show results where the treated individuals instead consist of individuals that developed mild side effects from pandemrix. These side effects are mostly self-reported and examples include fever, connective tissue pain and headache. The coefficients are much smaller than in Figure 7 but still meaningful. The fact that they developed mild symptoms is unlikely to change their belief about the risk of severe side effects in itself. Instead, we hypothesize that they more easily recall the experience of developing narcolepsy from pandemrix and therefore become more vaccine-pessimistic.

Learning about predisposition The finding that direct family members have similar but weaker effects on immunization outcomes as illustrated in Figure 7 may be partially explained by close family members sharing genetic traits with diseased individuals, leading them to perceive that they possess predispositions that make them more susceptible to side effects from vaccination. In Figure C1, we show results for the partners of the diseased individuals. For partners we find large estimates within the size range of those found for diseased individuals and close family members. Keeping in mind the endogeneity in partner choice, we interpret these findings as evidence that the results in Figure 7 for network members are primarily driven by information about the risk of severe side effects, rather than by learning about one’s predisposition to side effects.

Mechanisms alternative to altered perceived costs Our simple conceptual framework suggests that treated individuals are primarily influenced by updated beliefs about the risk of side effects. The framework also allows for the theoretical possibility that they respond to changes in perceived infection risk, or, put differently, to altered beliefs about the benefits of vaccination. To explore the possibility that decreased expected benefits of vaccination explain the worsened immunization outcomes, we elicit the perceived benefits of vaccination. We assess treated individuals’ concern about the disease through two different measures: (i) the number of COVID-19 self tests that they take, and (ii) whether they make phone calls for medical advice related to COVID-19 during the

pandemic. Because of differences in practices across regions regarding both testing and medical advice, we match treated individuals to individuals living in the same municipality. In doing so, we select control individuals from the overall Swedish population rather than from individuals who received pandemrix.

The results are displayed in ???. If anything, diseased individuals are less concerned about developing COVID-19 both measured by the number of tests and number of phone calls. The coefficients are small in magnitude but suggest that the main results may be partly explained by differing assessments of the risk of contracting COVID-19 or developing severe symptoms from COVID-19, rather than solely by the perceived risk of side effects. A simple story that we do not rule out, consistent with these findings, is that treated individuals do not expose themselves as much to other individuals, thereby reducing their risk of contracting COVID-19.

Experiencing the benefits of vaccination We augment the analysis by focusing on individuals who directly experience the benefits of vaccination, rather than its costs. The vaccination for influenza provides a suitable example to test this. Influenza is a highly infectious, virus-induced illness that is commonly known and has severe to even fatal medical symptoms. Annual immunization is recommended for the general public and has been shown to be a safe and effective strategy for reducing both the incidence and severity of seasonal flu outbreaks. Individuals developing severe cases of influenza, are most aware of the consequences of foregoing the annual flu shot. In the context of COVID-19, the memory of the benefits of vaccination may trigger a more positive response to vaccination. In Figure C3, we consider the same outcome variables as previously, but where the independent variable now takes the value 1 if an individual has been hospitalized for influenza at some point between 2015 and 2020. The control group are matched individuals selected from the universe of individuals living in Sweden in 2021. Our results provide suggestive evidence that the exposure to the benefits of vaccination reduces future vaccine hesitancy. Individuals receiving specialized healthcare for influenza may be selected in a variety of ways. We make progress in establishing causality by comparing the effects of developing influenza to the effects of developing similar diseases that are known to be not preventable by vaccines, namely upper respiratory tract infections (URTI). These diseases have similar symptoms and are, by definition, similarly severe in the sense that they all lead to hospitalization. We restrict our attention to hospitalizations where the primary reason for hospitalization was influenza or URTI. Overall, we find that individuals who experience influenza

have higher vaccination rate, take more doses and take their first dose earlier than comparable individuals. For URTI the effect is not as clear-cut. Individuals exposed to severe cases of URTI have a significantly lower up-take of COVID-19 vaccines, if anything slightly higher number of doses, and vaccinate significantly earlier than individuals in the control. Under the assumption that the selection bias is the same for influenza and URTI, we interpret the large difference between the coefficient for influenza and URTI as the causal effect of exposure to vaccination benefits on the COVID-19 outcomes.

Behavioral Explanations Up until now we have provided little room for behavioral explanations, different from the cost-benefit analysis, for the effects that we find. One such behavioral mechanism is that individuals identify more strongly with the affected person if they are similar, and thus react more strongly. We explore this hypothesis in Figure C4 by comparing cousins who share the same gender and age as the individual with narcolepsy to those cousins who do not. We find no evidence of larger effects among those who are more similar.

Effects beyond family members We proceed by providing results for extended, larger, networks. The goal of this exercise is to gain suggestive insights into the large-scale impact of the narcolepsy health scandal. If personal networks are permeable to the effects of exposure, extending beyond close family, this may indicate that aggregate vaccination uptake is likely to be lower in the wake of side effects scandals. Since most of the individuals were in school when they developed narcolepsy, we focus on schoolmates and neighbors in 2011, during the time the scandal unfolded and most of the individuals started experiencing symptoms. For colleagues we instead restrict our attention to individuals that were colleagues (working at the same plant) around the time of COVID-19 vaccination, in 2021. These networks are large, where treated individuals have on average 300 schoolmates, 33 colleagues, and 1,528 neighbors. Note that while these networks are constructed to enable exposure to the diseased, it is not certain that network members know a focal individual, and even if so, they may be unaware of their narcolepsy diagnosis. Figure 9 show results for colleagues, schoolmates, and neighbors. We find no indication that vaccine hesitancy spreads among these extended networks, as measured by either of our indicators of COVID-19 vaccine hesitancy.

[Figure 9 about here]

Attributing other diseases to pandemrix As stated in the data section, the best available evidence suggests that Pandemrix did not cause other diseases than narcolepsy. However, individuals may still attribute other diseases to Pandemrix, particularly if there were prior concerns linking the vaccine to that disease. We examine this by considering a number of diseases that were studied soon after the swine flu vaccination campaign (Persson et al., 2014). In Figure C5 we display COVID-19 vaccination rates as a function of the year in which it was first diagnosed for these different diseases. Since these diseases are much more common than narcolepsy, the data allow us to restrict the analysis to individuals who have received pandemrix. In short, we find no indication that individuals attributed their disease status to pandemrix, once again highlighting that individuals are rational in the sense that they only react to side effects that were confirmed to be associated with narcolepsy.

Healthcare visits Finally, we examine if treated individuals refrain from the healthcare system altogether. In Figure C6 we display the yearly number of specialist healthcare visits across time. In a short period after developing narcolepsy patients make more visits to medical professional, likely in examination phase, before they are diagnosed with narcolepsy, after that their number of non-narcolepsy related healthcare visits reaches a level comparable to the control group.

Taking stock Affected individuals and their family members display a large difference in COVID-19 immunization outcomes. We find similarly large effects among partners of diseased individuals, suggesting that individuals learning about a potential pre-disposition to side effects is not a key driver of the results. A theoretical possibility that we rule out is that results are explained by differing assessment of the benefits of vaccination. Past exposure exposure the benefits of vaccination as measured through individuals that experience influenza, rather than the costs, leads to improved immunization outcomes. We find little effects beyond family members, providing suggestive evidence that the scandal had limited impact on overall COVID-19 vaccination rate. A central question is why individuals put so much weight on their own experiences. (Graeber et al., 2024) provides evidence of stark differences in the ability to recall information that are conveyed through stories as opposed to statistics; stories are more vivid and easier to recall. In our setting, the exposure to narcolepsy may impede the ability to recall relevant statistics about risks of side effects.

7.2 Heterogeneity in Size of Database & Health Literacy

We now test two implications of our conceptual framework as laid out in Section 3. Prediction 1 states that the immunization response should be stronger among individuals with low health literacy, whereas Section 3 posits a stronger response among those with fewer prior healthcare experiences—that is, a smaller experience database. We focus on *Vaccine Taken* for individuals that develop vaccine side effects; results are similar for non-vaccine side effects, narcolepsy, and other outcome variables.

[Figure 8 about here]

In Figure 8, panel A we examine the role of the size of the database. We directly measure the size of the database, i.e. the number of experiences that could both influence an individual’s decision to receive the COVID-19 vaccine and interact with experiences of adverse events. We do this in two ways: (i) The number of unique drugs, (ii) The number of healthcare visits. Individuals who have taken a greater number of unique drugs or have more previous healthcare visits, and hence have a larger database, do not react less to a severe general side effect. This is surprising since those who have more experience with different drugs should update their belief about the risk of side effects to a lesser extent than those with fewer experiences. In Figure 8, panel B we examine the role of health literacy which in turn governs how much individuals rely on official information (θ in our model). We use three different proxies for health literacy: whether individuals (i) have a doctor in the family, (ii) have above-median cognitive ability, and (iii) have a university degree. Taken together, the estimates indicate that higher health literacy does little to dampen the behavioral response: better-informed individuals are no less likely to reduce their immunization uptake after an adverse event. A similar pattern emerges in Malmendier et al. (2021) who shows that even central bankers, arguably a group with expertise in inflation forecasting, let their own inflation experiences shape their forecasts. Archibong and Annan (2023); Anderberg et al. (2011); Chang (2018) likewise find larger effects on vaccine hesitancy among individuals with higher education after exposure to medical scandals, plausibly because greater information access heightened the scandal’s salience.

8 Exposure to Other Severe Side effects

Up to now, we have focused on the effects of exposure to narcolepsy. This was a particularly severe side effect, and the deployment of the swine flu vaccine shared remarkable similarities with the deployment of the COVID-19 vaccine. Moreover, we have strong reasons to believe that developing and reporting narcolepsy as a side effect is plausibly orthogonal to factors influencing vaccine hesitancy, as narcolepsy has few comorbidities and typically requires medical attention. We now consider the effects of severe side effects from all pharmaceutical products (including all vaccines apart from pandemrix).

Unlike the rare, vaccine-specific episode of pandemrix-induced narcolepsy, the focus on serious adverse events that recur across drugs, countries, and time allows us to address the policy-relevant question of how routinely experienced side effects shape future vaccine hesitancy. Because Swedish healthcare professionals are expected to report all suspected adverse events, regardless of whether the symptom is already recognized as vaccine-related, the national register captures a broad and representative spectrum of adverse reactions. By focusing on events classified as serious and reported by physicians, our analysis addresses reactions often severe enough to necessitate hospitalization, those most likely to shape subsequent immunization decisions.

we focus on 1 700 individuals reporting vaccine side effects and 33 000 individuals reporting non-vaccine side effects between 01 January of 2014 and 31 December of 2020—right before vaccination against COVID-19 started in Sweden. We focus on the effect of *severe* side effects that are reported by healthcare professionals. Individuals developing and reporting side effects differ from those that do not in a variety of ways. In Figure 10 we display how individuals reporting side effects differ along a few socioeconomic and health characteristics along with the differences in a matched sample. Individuals reporting side effects appear to be generally sicker, as they reveal a greater number of sick leave days, healthcare visits and drug prescriptions, but do not appear positively selected on socioeconomic characteristics.⁹ To maximize the number of observations we impose no restriction on the individuals in the sample having taken pandemrix. This, however, means that we can directly assess how individuals reporting side effects differ in their propensity to vaccinate by considering their vaccination rate with pandemrix. Figure 10 reveals that individuals that later

⁹Individuals that report mild side effects are, however, positively selected in terms of income and years of schooling, etc..

reported side effects were *more* likely to take the pandemrix vaccine. This is strong evidence that our results—both on pandemrix-induced narcolepsy and on non-vaccine side effects that we are providing in this section—can not be explained by latent vaccine hesitancy that correlates with both the propensity to report side effects and willingness to take the COVID-19 vaccine.

We identify a control group of comparable individuals in two steps. First, we find candidate control individuals who took the same drug the same year and who share the same birth year. Within these *birth year* $\times$ *drug* $\times$ *drug year*-cells we then perform nearest neighborhood matching where propensity scores are computed using a recurrent neural network based on socioeconomic and health variables.

[Figure 10 about here]

The individuals that report non-vaccine side effects and vaccine side effects are relatively older than those that developed narcolepsy. In 2021, when the vaccine was rolled out, the median age of the individuals that reported side effects was 63. At this age, forgoing or delaying COVID-19 vaccination entails substantial health risks compared to the average individual that developed narcolepsy.

What does experiencing a side effect entail? Figure C2 shows health outcomes around the time of reporting a side effect relative to a matched control group. Experiencing a side effect is associated with worsened health outcomes along observable measures: on average, affected individuals have three additional visits to specialized healthcare and twelve more days of sick leave in the year of experiencing a side effect. These effects also extend beyond up to two years after the event year. Importantly, when we restrict the attention to side effects from vaccines, the effects on health are roughly the same. This implies that any differences in effects between vaccine side effects and non-vaccine side effects do not stem from differences in severity.

We now turn to the results for non-vaccine side effects and vaccine-related side effects. These results are displayed in Table 1.

[Table 1 about here]

For diseased individuals, the broad pattern is that these individuals become more hesitant with respect to the COVID-19 vaccine as measured by our three immunization outcomes. Interestingly, the coefficients are roughly proportional to the results that we find for narcolepsy with the coefficients for vaccine taken and for days elapsed being about 1/20 for non-vaccine side effects and about 1/3 for vaccine side effects of the coefficients that we find for narcolepsy. For family members we find

precisely estimated null results for non-vaccine side effects but meaningful effects for vaccine side effects. Recall that there is no meaningful difference in severity between non-vaccine side effects and vaccine side effects. Instead, the larger effects for vaccines stem from the perceived similarity to COVID-19 vaccines which boosts both recall and simulation compared to non-vaccine side effects. One concern regarding the argument of similarity is that individuals reporting vaccine side effects are, on average, younger (mean age = 39, SD = 25) than those reporting other side effects (mean age = 55, SD = 20). Since younger individuals face lower health risks from remaining unvaccinated, this age difference may affect the comparability of the groups. In Table C2 we reweight observations in the vaccine sample to match the age distribution of individuals in the non-vaccine side effects-sample. The coefficients decrease only marginally.

Stickiness of experiences In ?? we break up the results by year of reporting side effects between 2013 and 2020. There is no indication that the effect decays over time. While the side effects are not on top of individuals’ mind, the memories are reactivated years later once facing the decision to vaccinate against COVID-19. A plausible concern is that there is an interaction between the salience of side effects and developing a side effect. In other words, individuals that developed side effects shortly after the narcolepsy scandal unfolded are more likely to be affected. The salience interaction could explain the dip in uptake among those who experienced vaccine side effects in 2016, when public debate over narcolepsy compensation resurfaced as the government assumed responsibility for compensating affected individuals.

Learning about predisposition If individuals learn about a personal predisposition to side effects after experiencing one, abstaining from future vaccines can be rational. We classify drug–side-effect pairs as either *idiosyncratic* or *type-learner* using a large language model. It is very hard to establish, for each drug-side effect combination, whether an individual is more likely to experience a side effect from the COVID-19 vaccine. The point of using a LLM is not only that it is feasible to classify a large set of combinations but also that it broadly reflects expert knowledge and hence the perception of the affected individual regarding whether a drug side effect-combination is idiosyncratic or not. We construct a binary indicator that splits treated individuals 50/50 across the two classes, maximizing power while remaining agnostic about the exact degree of predisposition. In regressions of subsequent vaccination, the effect is 0.147 (SE 0.028) for type-learner side effects and 0.0679

(SE 0.023) for idiosyncratic side effects; the difference is 0.079 (SE 0.036), $p \approx 0.03$. These results suggest that individuals indeed react significantly more to side effects where they are likely to learn that they have a predisposition to side effects.

9 Spillover to Children’s Vaccines

We proceed to examine the effects of exposure to side effects on vaccines against diseases other than COVID-19. In particular, we focus on children’s vaccines against measles, mumps and rubella (MMR), diphtheria and pneumococcal disease. The purpose of this exercise is twofold.

First, we assess whether the effects we observe for the COVID-19 vaccine are specific to newly introduced pharmaceutical products. Our conceptual framework proposes that, because external information is sparse in unprecedented situations such as COVID-19, individuals rely heavily on their own prior experiences. The opposite holds for established drugs, such as common child vaccines, which have historically low and well-documented incidences of severe adverse events. The diphtheria vaccine has been included in the program since the 1940s, and measles—via the MMR vaccine—since 1982, with a standalone measles vaccine available from 1972. The pneumococcal vaccine was added to the Swedish program in 2009, though available in the US since 1977. In consequence, easy access to scientific evidence lowers information costs and increases reliance on external sources. Against this background, we interpret any change in vaccination outcomes from adverse events related to established child vaccines as suggestive evidence of a shift in the degree to which individuals rely on their own experience θ vis-à-vis best available evidence.

Second, we explore whether the effect of side effects on vaccine outcomes translate into situations where health decisions are made on behalf of others and individuals do not bear the immediate consequences of their decision. Parents have a mandate over health care decision of their children from an early age and thus play a crucial role on whether the children ever come in contact with the public healthcare system throughout their childhood.

We define a sample of individuals born in 2013 and onward who are registered as living in Sweden throughout their entire lives. The three standard vaccines we focus on are administered for the first time below two years of age. This includes the measles vaccine, first administered at 18 months; the diphtheria vaccine, first due at 3 months, and most often administered in combination with polio, tetanus, Hepatitis B, pertussis, and influenza type B; the pneumococcus vaccine, which is also

due at 3 months of age. Both pneumococcus and diphtheria are on average first administered at approximately 3.2 (SD=1.74) and 3.7 (SD=5.73) months of age in our data set. The average first vaccination age for the measles shot is in line with administrative regulations showing an average age of 19 (SD= 12.02) months. Our adherence shares resemble official numbers by the Swedish public health authority (Folkhälsomyndigheten, 2024). A share of individuals abstain from taking the vaccines in the standard children’s vaccine program altogether. Notably, among 4.1% of children, parents abstain from having the measles vaccine administered, compared to 1.3 % for diphtheria and 2 % for pneumococcus. The risk of developing any of these diseases for unvaccinated individuals is very low. Instead, a plausible explanation is the perceived link between measles and autism (Stratton et al., 2001) that we will investigate later in this section. Panel B of Table 2 illustrates that children vaccination decisions correlate highly. This observation suggests that parental vaccine hesitancy reflects a general sentiment rather than concerns about a specific vaccine.

[Table 2 about here]

9.1 Exposure to Narcolepsy

In a first step, we study the children of individuals that themselves developed narcolepsy after receiving Pandemrix. As our prior results indicate that effects propagates within families, we also extend the analysis to nieces and nephews. The dependent variable equals one if a child completed the vaccination schedule—defined as receiving one dose of the MMR vaccine and three doses each of the pneumococcal and diphtheria vaccines—by age two. Table 3 displays results for children in panel A and nieces and nephews in panel B of individuals that developed narcolepsy and matched control individuals. Column 5 contains the estimates for an aggregated index of adherence, defined as the average vaccine adherence for the three different diseases examined. Given our data limitations, we are restricted to a few individuals in the treatment group who have children born after 2013, which means that our standard errors are large. We deploy a simple matching strategy where we match exactly on coarsened versions of socioeconomic characteristics of the parents as well as birth month. The data does not allow to observe if the children get the vaccine later than what is scheduled. Furthermore, we focus on the first difference τ_{post} , and therefor do not remove potential effects from being a network member of someone who developed narcolepsy prior to the swine flu pandemic. Since we regress a binary outcome variable on a binary treatment variable we augment the OLS with odds ratios along with confidence intervals computed using and confidence interval obtained

by inverting Fisher’s exact test. This avoids reliance on asymptotic normality for the odds-ratio. Against this background, the negative point estimates are large, indicating that individuals who develop narcolepsy also abstain from fully vaccinating their children with well established vaccines.

We interpret these results as suggestive evidence that trust in information regarding the safety of vaccines provided by healthcare authorities and scientific authorities plays a mediating role in vaccine hesitancy. Revisiting our main results against this knowledge, an increase in vaccine hesitancy observed in lower COVID-19 vaccination rates, delayed first vaccination contact and incomplete vaccine schedules, could be induced by a shift in θ . As illustrated in Panel B, we find no similar effects for nieces and nephews, indicating once again that the impact of vaccine-induced narcolepsy on overall immunization outcomes is limited with regard to personal networks.

[Table 3 about here]

9.2 Exposure to General Adverse Events

In line with our analysis of narcolepsy vis-à-vis other adverse events in the previous sections, we now consider the effects from other severe adverse events. Once again, We consider the universe of adverse events, excluding any reports from pandemrix and COVID-vaccines. Importantly, the longitudinal data structure allows us to account for unobserved time-invariant vaccine hesitancy by exploiting the variation of adverse events across children within the same family. To this end, we estimate the following regression equation:

$$y_{pb} = \delta_b + \gamma_p + \beta T_{pb} + \epsilon_{pb} \quad (6)$$

Where δ_b denote birth order fixed effects, γ_p denote parent fixed effects and T_{pb} is equal to one for a child with birth order b is born to a parent that reports a severe side effect, after the parent has reported a side effect. y_{pb} is the vaccination status for the child. We construct sibling groups based on each parent and assign weight 0.5 for each child that appears twice so that each child contributes equally to the estimate regardless of whether one or two parents are observed. Standard errors are clustered by parent.

[Table 4 about here]

For general side effects we find no negative effects on vaccine uptake among children’s vaccines for children born after a parent reports a side effect. In particular, the estimates for the MMR vaccine and the pneumococcus vaccine are close to zero. When we restrict the analysis to side effects

from vaccinations we find negative estimates that are statistically significant for the MMR vaccine and close to significant for the vaccine against pneumococcal disease. Although we lack statistical power, these findings, together with the findings for children of individuals with pandemrix-induced narcolepsy, suggest that experiencing adverse events from vaccines does cause individuals to be more reluctant vaccinating their children with well-established and safe vaccines.

10 Fact vs Fallacy - The case of Autism

Finally, we aim to understand whether adverse events need to be backed up by scientific evidence to produce the substantial negative effects on immunization outcomes observed in the previous sections. In particular, we isolate a scenario where an adverse event is likely perceived to result from medical treatment but lacks scientific recognition. To this end, we exploit an infamous scandal surrounding the MMR vaccine and study younger siblings of children who developed autism soon after receiving the measles vaccine.

In 1998, a controversial study of 12 children was published in *The Lancet*, suggesting a potential link between the measles, mumps, and rubella (MMR) vaccine and bowel disease, as well as autism. Based on the article, the lead author Andrew Wakefield advocated against continued use of the triple MMR vaccine in the popular press. Media coverage at the time played a major role in disseminating the suspected link to the public. A 2004 journalistic investigation exposed a serious conflict of interest, as the lead author profited from the demotion of the MMR vaccine (Deer, 2020). The article was retracted in 2010 after the General Medical Council revealed data manipulation and nontransparent case selection. Since then, several large-scale studies have found no causal association between the MMR vaccine, bowel disease, and autism. Nevertheless, as shown in table 2, the publication had a lasting impact on public perceptions of vaccine safety. Motta and Stecula (2021) find that aggregate vaccine hesitancy towards the MMR increased, as parents reported more negative events related to the vaccine following the Wakefield article.

In contrast, we rely on personal diagnosis history to identify individuals that develop autism at a young age. Similar to the previous section, we exploit within-family variation by comparing children born before and after an autism diagnosis in the same family. The results are presented in Figure 11. We display results for autism together with results for other mental and behavioral disorders (diagnoses ICD-10 chapter V) and other common diseases developed before the age of age

5. Our findings suggest that children with an older sibling with autism are 4 percentage points less likely to receive the measles vaccine (but not other common childhood vaccines). A few connections to our previous analysis can be made. Once again, the effects are domain specific in the sense that parents do not abstain from the diphtheria vaccine or the pneumococcal vaccine. We find a similar suggestive pattern for mild intellectual disability, which arguably shares several features with the symptom profile of autism. Note, that we study a sample of individuals well after the controversial article linking the MMR-vaccine to autism was retracted in 2010. Our results suggest that even after the retraction of the Wakefield article in 2010, many parents continue to attribute autism to the measles vaccine. This highlights that controversial side effects can persist in public memory irrespective of their scientific foundation: what matters is not scientific consensus, but the presence of a salient cue linking a treatment to a symptom. In line with our broader framework, families exposed to severe side effects appear to discount official information and instead overweight their own experiences.

11 Discussion

In this paper we provide evidence that adverse drug events affect future healthcare consumption. We use novel and rich Swedish register data on adverse drug reactions and vaccine uptake to measure how narcolepsy, a severe neurological disease linked to the swine flu vaccine in 2010, affected immunization choices during the COVID-19 pandemic. We find that exposure to the side effect of narcolepsy leads to significantly lower COVID-19 vaccine take up, uptake of fewer doses, and delay in vaccination for the vaccinated diseased and their close family members. We attribute these findings on vaccine hesitancy to an increase in expected costs of vaccinations, in particular through updating of the perceived risks of developing a side effect. Taken together, the effects of experiencing adverse events are sticky in that they affect the diseased individual a long time after experiencing an adverse event, but are specific in that the experiences only inform a narrow set of similar future decisions that the diseased individual face. Exploring heterogeneity in our results, we find remarkable little differences in effects across different subgroups which leads us to conclude that improving health literacy does not necessarily mitigate the effects on vaccine hesitancy.

The results on autism and measles vaccinations suggest that public health authorities needs to be cautious about the communication of the risk adverse drug events, as false narratives may

become part of public memory even if they are not backed-up by scientific evidence.

While previous economic research on medical scandals has largely focused on rare and extreme cases we examine ordinarily occurring vaccine side effects, providing evidence with immediate relevance to drug safety regulation and vaccine communication strategies. Our results suggest that it is necessary for policy makers to account for the additional cost stemming from future decreased health care decisions when introducing new medical treatments. If medical treatment are perceived to be associated with severe side effects, this may have long-lasting negative effects on the uptake of similar new health care products but also the demand for well established health products, which in turn bears large social externalities.

This paper contributes to a growing literature in behavioral economics on experience-based decision making. In particular, we study the role of previous experiences in a high-stakes decision where both personal health and others' health are at stake.

An interesting venue for future research is to explore the role of informational nudges that provide individuals that have experienced severe side effects with the true frequency of side effects. Future research should aim at exploring the extent to which individuals choose not trust publicly provided information on the safety and efficacy of vaccines and ultimately better quantify the extent to which severe side effects erode public trust in scientific and health authorities. In a similar vein, another interesting venue is to explore whether public recognition and reimbursement serves to alleviate the negative effects on vaccine uptake by restoring trust among affected individuals.

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12 Tables and Figures

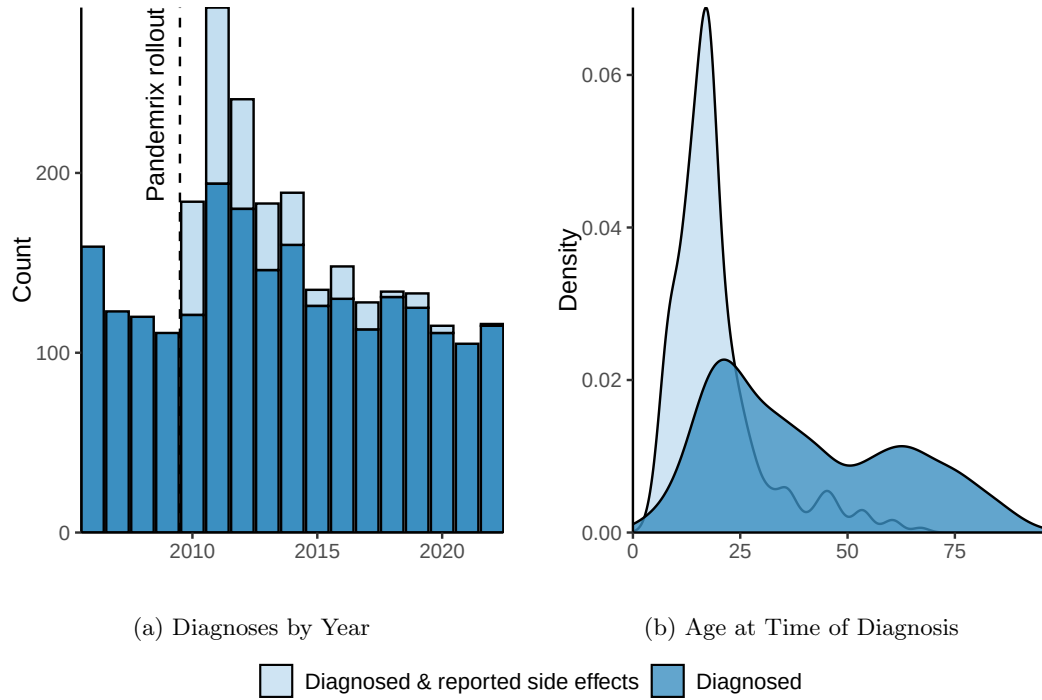


Figure 1: Timing and Age Distribution of Narcolepsy Diagnoses, by Reporting Status

Notes: This figure displays diagnosis characteristics for individuals that reported narcolepsy as a side effect to the Medical Products Agency in dark blue and those that were diagnosed but did not report it as a side effect in light blue. **(a)** Count of individuals receiving a first narcolepsy diagnosis in specialized healthcare by calendar year. **(b)** Density of age at first narcolepsy diagnosis.

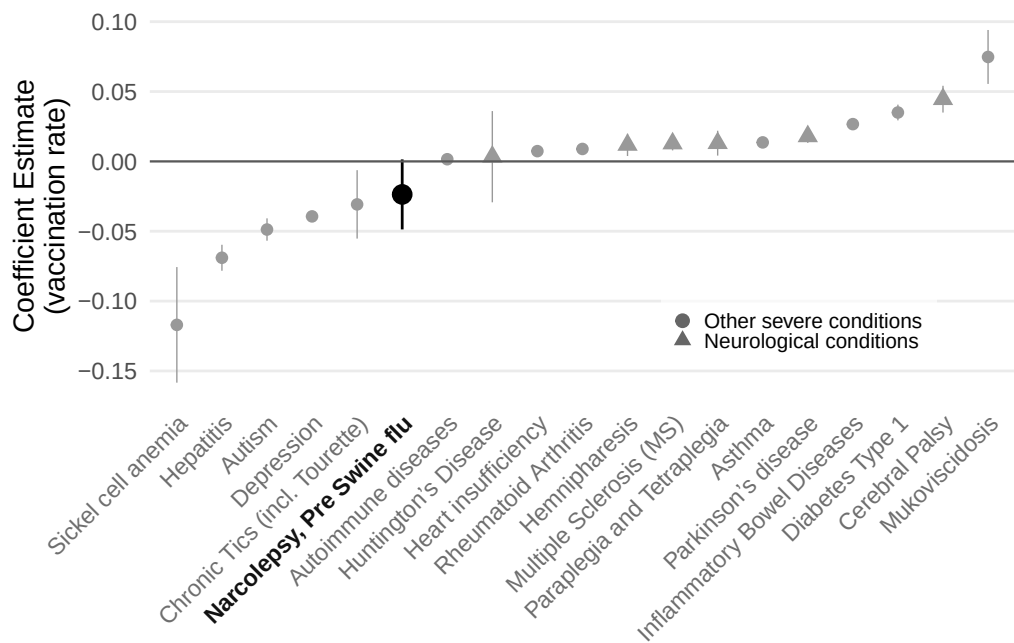


Figure 2: Saliency of Pandemrix Induced Narcolepsy.

Notes: This figure displays estimated differences in COVID-19 vaccination rates between groups of individuals with different diseases. The control group for all regressions is a random subset of 1 000 000 individuals living in Sweden in 2021. All regressions include fixed effects for birth year and gender. *Narcolepsy, pre swine flu* refers to individuals that developed narcolepsy before vaccination with Pandemrix began.

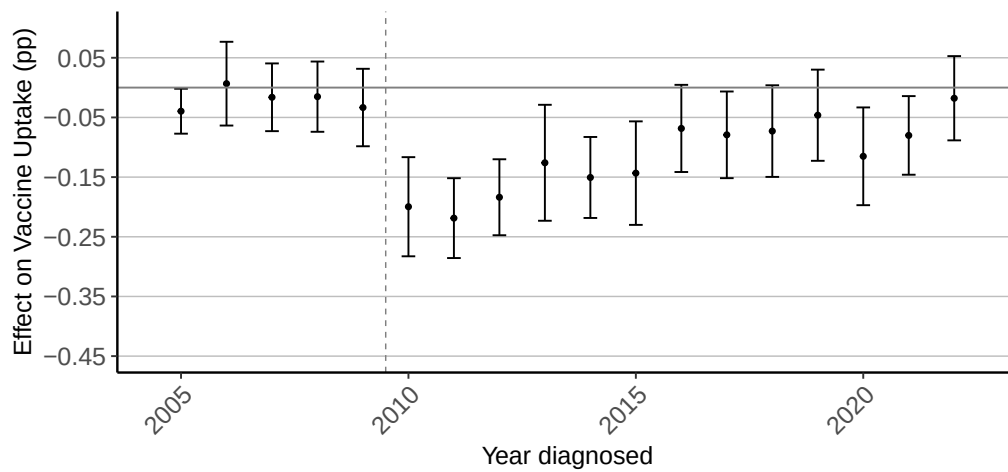
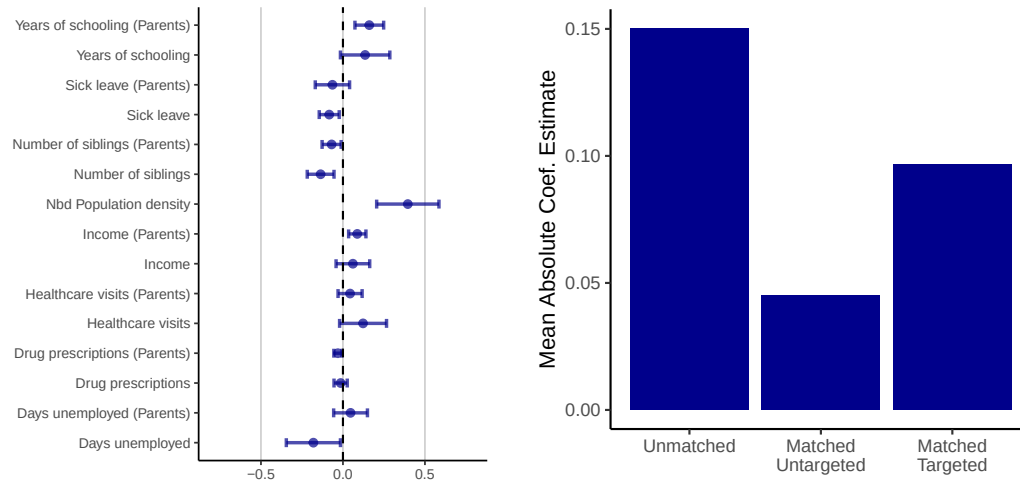


Figure 3: Effect on COVID-19 vaccine uptake of Narcolepsy Diagnoses Across Time

Notes: Differences in COVID-19 vaccination rates for different years of first being diagnosed with narcolepsy. The control group for all estimates is a random subset of 1,000,000 individuals living in Sweden. The vertical dashed line marks the onset of pandemrix vaccinations. All regressions include fixed effects for birth year and gender.



(a) Coefficient Forest Plot

(b) Aggregate Imbalance before/after matching

Figure 4: Pre-Treatment Covariate Balance

Notes: **(a)** Differences in a selection of pre-determined characteristics between treated and pandemrix control individuals. Each estimate comes from a univariate regression of the standardized covariate on the treatment status and on birth year fixed effects. Own socioeconomic characteristics, as opposed to parental characteristics, are restricted to individuals born before 1990. **(b)** Mean absolute value of coefficient for all covariates used. The first column displays the mean value before matching. The second column displays the mean value for variables not selected for matching. The third column displays the mean value for variables selected for matching.

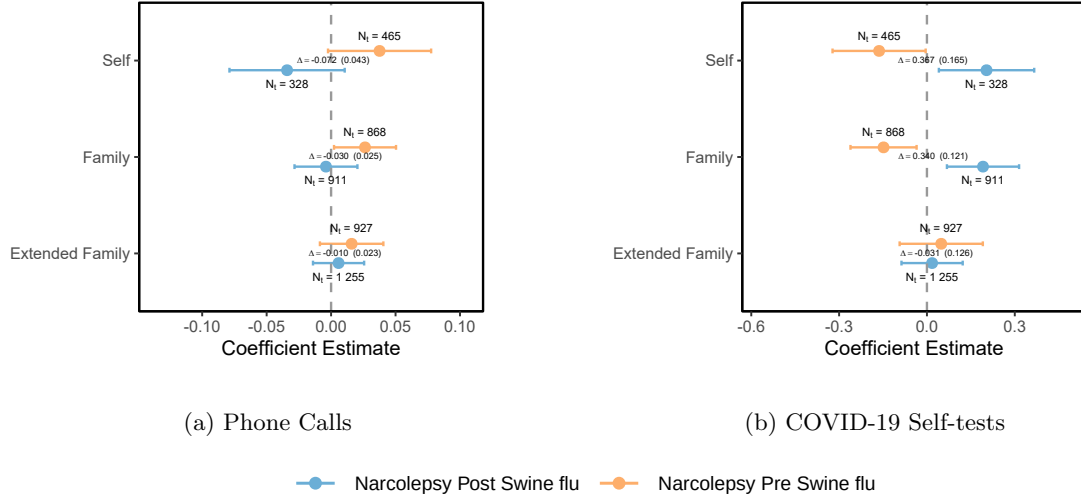


Figure 5: Eliciting the Perceived Benefits of COVID-19 vaccination

Notes: This figure displays coefficient corresponding to τ_{post} and τ_{pre} in eq. (4). First row corresponds to coefficient for individuals that themselves developed narcolepsy (Self). The second and third rows show the corresponding coefficients for Family members and Extended family members respectively. **Panel A:** Displays coefficient estimates for whether not an individual made a phone call for medical advice for COVID-19 symptoms during the pandemic. **Panel B:** Displays coefficient estimates for number of COVID-19 self-tests an individual has taken.

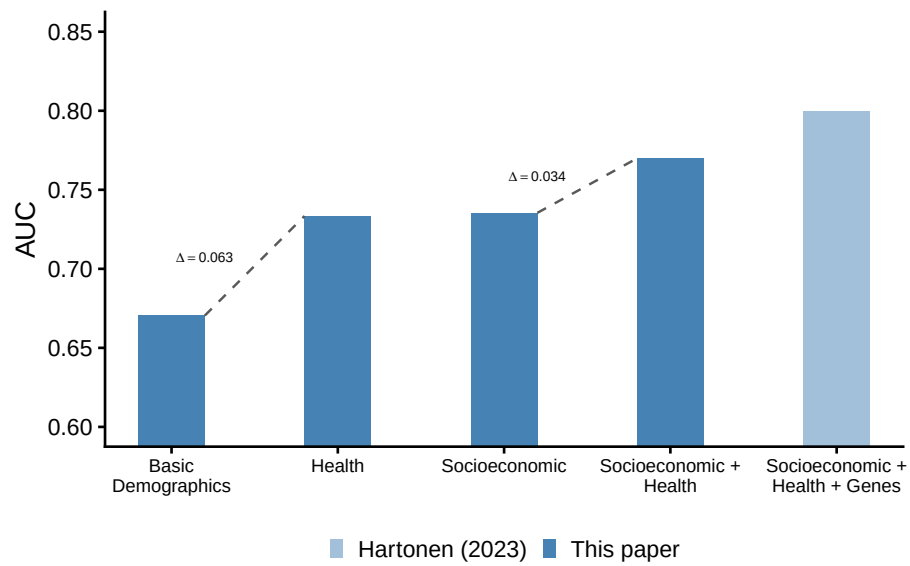


Figure 6: Predicting Who Takes the COVID-19 Vaccine

Notes: Out of Sample AUC-values for predicted probabilities using a neural network. Sample restricted to individuals between 40 and 60 years old who do not belong to COVID-19 risk groups. All models are evaluated out-of-sample. Models are described in details in F

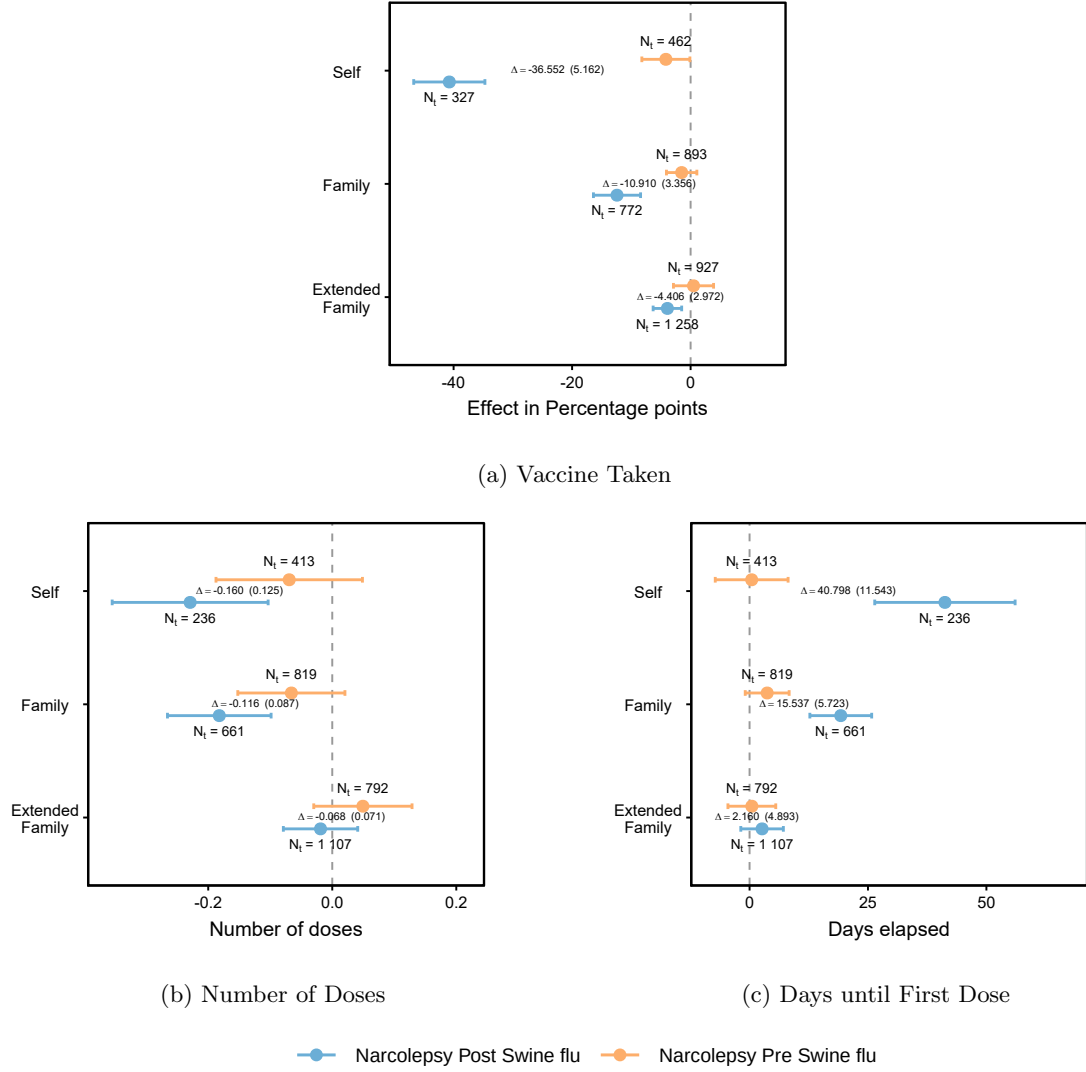


Figure 7: Main Results – Diseased Individuals and Family Members.

Notes: This figure displays coefficients corresponding to τ_{post} and τ_{pre} in eq. (4). First row corresponds to individuals that themselves developed narcolepsy (Self). The second and third rows show the corresponding coefficients for Family members, defined as siblings and parents, and Extended family members, defined as aunts/uncles and cousins. **Panel A:** Ever taken at least one COVID-19 dose. **Panel B:** Number of doses, conditional on at least one dose. **Panel C:** Days elapsed until the first COVID-19 dose. Standard errors are clustered by treatment cluster, i.e. by the treated or control individual that a network member is related to.

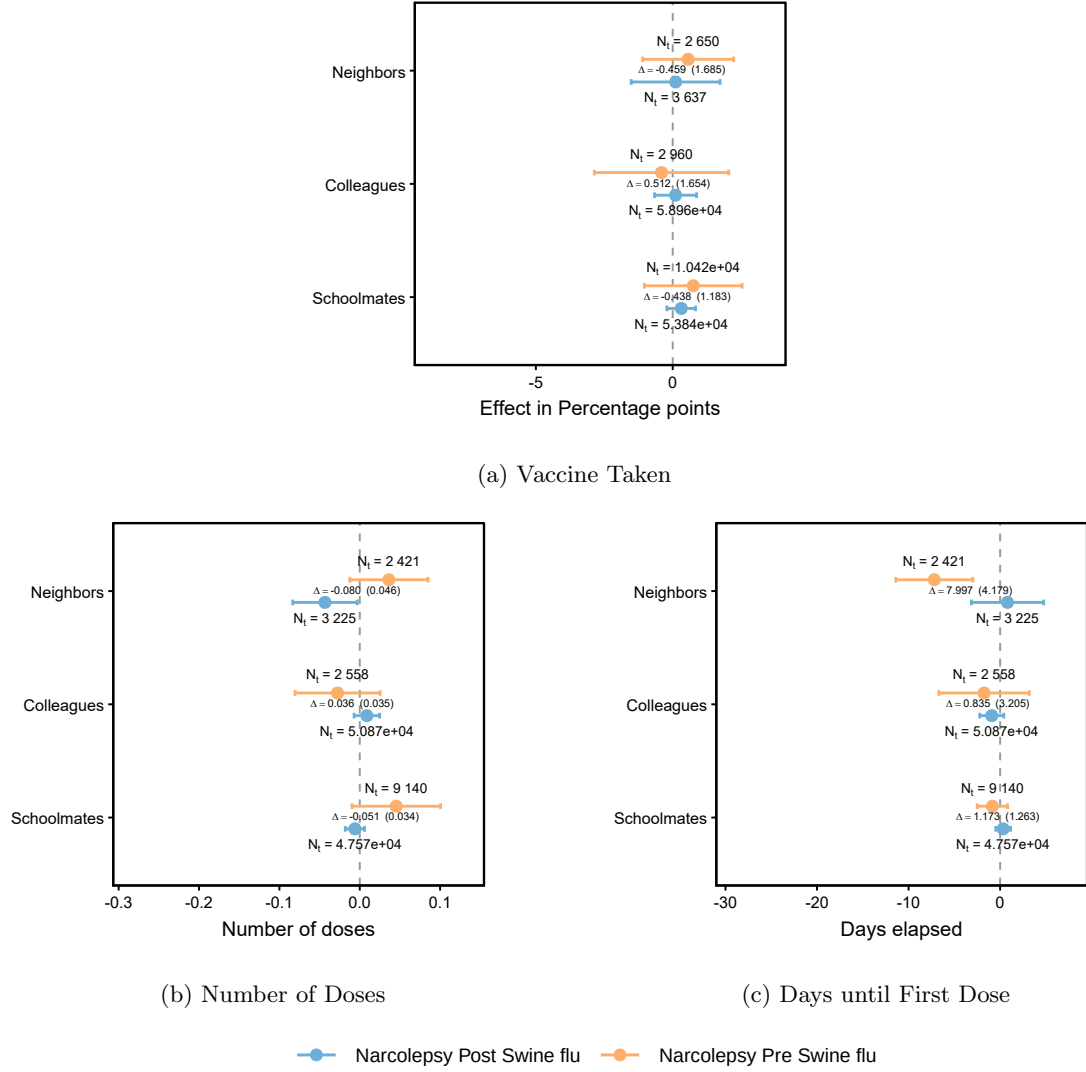
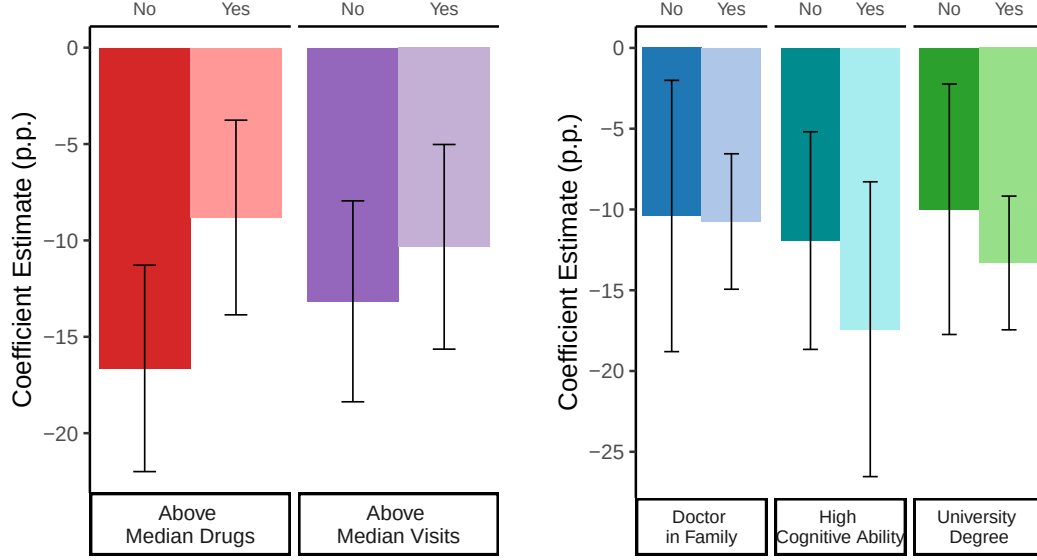


Figure 9: Main Results – Extended networks.

Notes: This figure displays coefficient corresponding to τ_{post} and τ_{pre} in eq. (4). The first row corresponds to coefficients for colleagues of core sample individuals in 2021. The second row corresponds to coefficients schoolmates to the sample individuals in 2011. The third row corresponds to coefficient for individuals that lived in the same neighborhood in 2011. **Panel A:** Displays results for whether or not an individual has taken at least one COVID-19 dose. **Panel B:** Displays results for the number of COVID-19 doses, conditional on having taken at least one dose. **Panel C:** Displays results for the number of days elapsed until the first COVID-19 dose is taken. Standard errors are clustered by treatment cluster, i.e. by the treated or control individual that a network member is related to.



(a) Size of Database

(b) Health Literacy

Figure 8: Testing Predictions From Model: Health Literacy & Size of Database.

Notes: This figure tests two of the predictions from the model: (a) The role of previous healthcare experiences and (b) The role of health literacy, which in our model manifests itself through θ . We restrict the attention to τ_{post} for family members and for the binary outcome variable *Vaccine Taken*. **(a)** *Above Median Drugs* is defined as "Yes" if an individual has above median number of drugs taken (≈ 6) between 2005 and the point of developing side effect relative to its birth year peers. *Above Median Visits* is defined equivalently but for visits to specialized healthcare between 2005 and 2009 (≈ 0.6). **(b)** *Doctor in family* is defined as "Yes" if an individual has a parent or a sibling with a medical degree or a nursing degree. *High Cognitive Ability* is based on cognitive tests completed by military conscripts. It is equal to "Yes" if an individual had above median score on the test compared to peers born the same year. *University Degree* is equal to "Yes" if an individual has at least a bachelor degree, corresponding to three years of higher education.

Table 1: Side Effects and Immunization Outcomes

	General Side Effects			Vaccine Side Effects		
	Vaccine Taken	Number of Doses	Days Elapsed	Vaccine Taken	Number of Doses	Days Elapsed
Diseased Individuals						
Side Effect	-0.006 (0.003)	-0.038 (0.015)	-3.19 (0.68)	-0.075 (0.019)	-0.056 (0.073)	10.3 (3.51)
N. Treated	12 230	11 326	11 326	672	553	553
Family Members						
Side Effect	0.002 (0.002)	0.030 (0.010)	0.30 (0.42)	-0.037 (0.011)	-0.092 (0.040)	5.91 (1.92)
N. Treated	28 509	26 323	26 323	1 643	1 431	1 431
Children						
Side Effect	0.009 (0.004)	0.021 (0.009)	-0.74 (0.56)	-0.042 (0.022)	0.056 (0.050)	7.42 (3.55)
N. Treated	24 188	19 670	19 670	669	513	513
Partner						
Side Effect	-0.001 (0.003)	-0.008 (0.020)	-1.32 (0.79)	-0.009 (0.015)	-0.056 (0.096)	7.90 (4.14)
N. Treated	6 216	5 951	5 951	245	237	237

Notes: This table displays results from regressing the COVID-19 immunization outcome variables on a variable indicating if someone developed an reported a side effect between 2015 and 2020. Columns 1–3 use an indicator for reporting any side effect between 2015 and 2020 as the regressor (“General side effects”). Columns 4–6 instead use an indicator for reporting a vaccine side effect (“Vaccine side effects”). Each treated individual is matched 1:1 to an untreated individual who (a) was born in the same year and (b) took the same drug in the same year as the treated individual.

Table 2: Childhood Vaccines – Descriptive Statistics

Panel A: Share adhering				
	Diphtheria	Pneumococcus	Measles	Covid (Parents)
Abstention	1.3	2	4.1	10.6
Partial adherence	4.6	9		31.0
Full Adherence	94.1	89	95.9	42.8

Panel B: Correlation matrix				
	Diphtheria	Pneumococcus	Measles	Covid (Parents)
Diphtheria	1.000			
Pneumococcus	0.739	1.000		
Measles	0.474	0.453	1.000	
Covid (Parents)	0.167	0.157	0.160	1.000

Notes: This figure displays the uptake of vaccines in the Swedish children vaccination program.

Panel A: Abstention is defined as having received zero doses of the vaccine. Partial adherence defined as having received one or two doses for diphtheria and pneumococcus. Full adherence defined as having received three or more doses for diphtheria and for pneumococcus and having received one or more dose for measles. **Panel B:** Correlations in having received at least one dose. COVID-19 (parents) is defined as the average of parental COVID-19 vaccination status.

Table 3: Results–Children’s vaccines

	Measles	Diphtheria	Pneumococcus	Index
Children				
OLS	-0.055 (0.038)	-0.041 (0.038)	-0.057 (0.047)	-0.051 (0.035)
Odds Ratio	0.612 [0.301, 1.409]	0.723 [0.329, 1.878]	0.652 [0.337, 1.382]	
Num. Treated.	64	64	64	64
Nieces and nephews				
OLS	0.008 (0.010)	0.016 (0.009)	0.007 (0.019)	0.010 (0.009)
Odds Ratio	0.727 [0.443, 1.264]	0.768 [0.457, 1.381]	0.853 [0.541, 1.406]	
Num. Treated.	157	157	157	157

Notes: This table displays results for the children’s vaccines for own children as well as nieces and nephews. The regressions are run on a matched sample where each child is matched exactly on birth month and where parents are matched on coarsened versions of parental years of schooling, education field, income, number of healthcare visits and number of drugs prescribed. The dependent variable is equal to one if an individual adheres to the vaccination schedule. Confidence intervals computed using Fisher’s exact test in square brackets.

Table 4: General Side Effects and Children's Vaccines

	Non-vaccine Adverse Events				Vaccine Adverse Events			
	Measles	Diphtheria	Pneumococcal	Index	Measles	Diphtheria	Pneumococcal	Index
Treated	0.009 (0.016)	0.018 (0.017)	0.006 (0.019)	0.011 (0.015)	-0.102 (0.050)	-0.028 (0.068)	-0.100 (0.063)	-0.077 (0.057)
Dep var. Mean	0.92	0.91	0.85		0.92	0.91	0.85	
Num.Obs.	2,054,933	2,054,933	2,054,933	2,054,933	2,054,933	2,054,933	2,054,933	2,054,933
R2	0.761	0.764	0.744	0.769	0.761	0.764	0.744	0.769
FE: Birth year	X	X	X	X	X	X	X	X
FE: Birth Order	X	X	X	X	X	X	X	X
FE: Parent	X	X	X	X	X	X	X	X

Notes: This table displays estimates of β from Equation (6). The dependent variable is a binary indicator equal to one if a child has received the recommended number of vaccine doses. Vaccine side effects refers to side effects reported from vaccines (ICD-10-SE J07), excluding pandemrix and COVID-19 vaccines. General side effects refer to side effects reported from other types of drugs. Standard errors are clustered by parent and child.

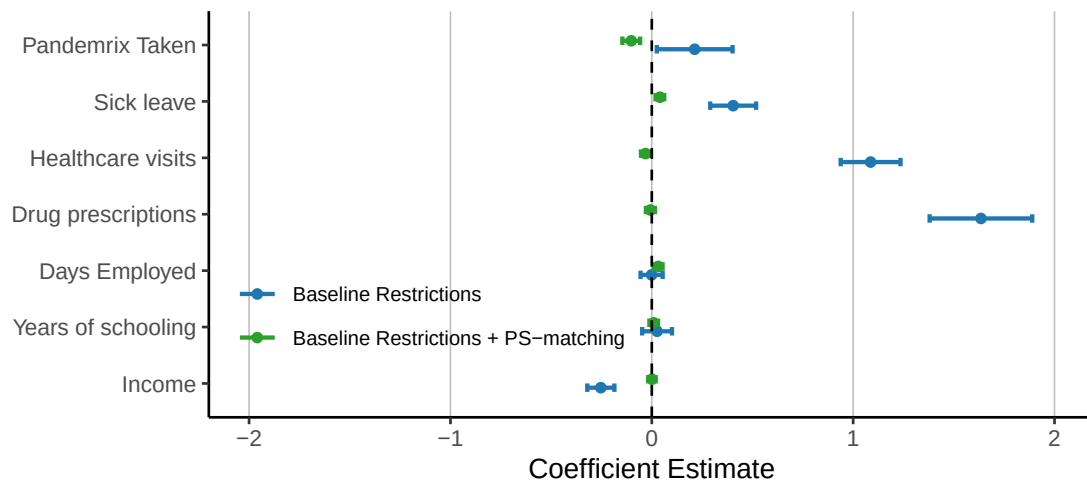


Figure 10: Balance – General Adverse Events

Notes: This table displays balance in general (vaccine and non-vaccine) adverse events before and after matching on propensity scores where the propensity scores are computed using a neural network. Each estimate comes from a univariate regression of the standardized covariate on the treatment status and on birth year fixed effects. Pandemrix taken is defined for a subset of individuals who lived in healthcare regions in 2009 where we access individual level data on pandemrix vaccination. Pandemrix taken is deliberately excluded from the computation of propensity scores as to display how well balance in immunization outcomes during the swine flu pandemic can be achieved using other health- and socioeconomic characteristics.

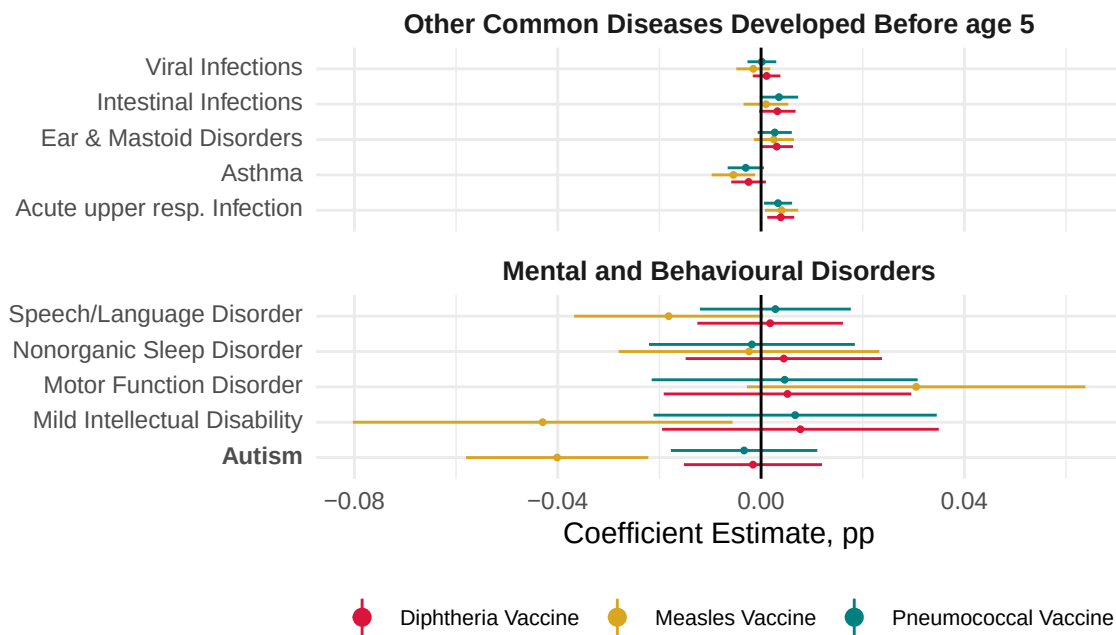


Figure 11: Vaccination Uptake After an Older Sibling Is Diagnosed With Autism

Notes: This figure presents estimated coefficients from regressing a binary vaccination adherence indicator on birth order and parent fixed effects, and a binary variable equal to one if the individual was born after an older sibling was diagnosed with autism or another specified disease. The sample consists of individuals born between 2013 and 2022. For autism, the number of treated individuals, i.e. individuals that are born after an older siblings vaccinates is equal to 6,347. Standard errors are clustered by parent.

Appendix A. Background

A.1 Narcolepsy

Narcolepsy is a chronic neurological disease that significantly disrupts the ability to control sleep-wake cycles. It is characterized by excessive daytime sleepiness (EDS), which is present in all individuals with the condition. Narcolepsy is classified into two types: type 1 and type 2, which differ in their associated symptoms. In type 1 narcolepsy, patients experience both EDS and cataplexy, a sudden loss of muscle tone triggered by strong emotions. This form is typically associated with an autoimmune process that leads to a deficiency in hypocretin, a neuropeptide critical to the regulation of wakefulness, in the central nervous system. In contrast, patients with type 2 narcolepsy exhibit EDS without cataplexy, and the severity of EDS is generally less pronounced compared to type 1 (see Baumann et al., 2014).

The clinical diagnosis entails detailed measurement of hypocretin concentration to assess the potential for type 1 narcolepsy. Furthermore, the patient will undergo lengthy examinations in sleep laboratories (multiple sleep latency tests) and several diagnostic tests for exclusion of other drivers of sleep disturbance (ie, analysis of blood plasma for iron deficiency). Due to the observability of cataplexies and established biomarkers available as diagnostic anchors, studies have found that type 1 narcolepsy to be detected more easily by medical practitioners.

Although there is currently no cure to the disease, treatment of symptoms using behavioral measures (i.e., sleep hygiene and schedule) and medical treatment (Barateau et al., 2016) is available. Patients with narcolepsy often develop conditions, such as metabolic and cardiovascular diseases, psychiatry diseases, musculoskeletal chronic pain, and other specific sleep disorders (Barateau et al., 2016).

The existing literature suggests that narcolepsy cases are associated with genetic predispositions, such as carrying the genom HLA-DQB1*06:02 and GDNF-AS1 (Hallberg et al., 2019; Gauffin et al., 2022). Around 30% of the Swedish and Finnish general population carry the respective alleles and therefore it is not sufficient to explain the development of narcolepsy (Partinen et al., 2014).

A.2 Legal process

Patients that developed narcolepsy post Pandemrix vaccine first claimed reimbursement from the Swedish Pharmaceutical Insurance, an insurance that is an agreement between almost all pharmaceutical companies operating in Sweden. The Swedish Pharmaceutical Insurance was obliged to pay out at most 150 million SEK (the common limit to payouts for all injuries attributable within one calendar year). "In 2016, the government decided to assume responsibility for compensating affected individuals who had not received adequate compensation from the insurance company, owing to its financial limitations. Individuals received at most 10 million SEK (900k USD) from the insurance company and the government combined - where the purpose of the payment is compensation for losses in salary. The insurance company and the government made different judgements on the requirements for receiving reimbursements but in general individuals aged above 20 at the time of the first vaccination and individuals who reported narcolepsy after three years of vaccination were less likely to receive reimbursement. Ten years after the swine flu vaccine campaign, some diseased individuals are still claiming and receiving reimbursements from the government for pain and suffering.

A.3 Swine flu pandemic and the vaccination campaign

In April 2009, the first cases of swine flu were discovered in Mexico. In June, three months later, the World Health Organization declared swine flu influenza a pandemic. Phase 1 clinical trials for Pandemrix were completed in September 2009, at which point Pandemrix was granted market authorization by the European Commission given the exceptional circumstances.

During the vaccination campaign, Sweden had the goal of reaching herd immunity. The general public was recommended to take the vaccine. However, some risk groups were prioritized including individuals with some chronic diseases, pregnant individuals, and healthcare workers, were recommended taking the swine flu vaccine. The vaccine was fully subsidized.

In total 60% of the population received the swine flu vaccine, which was lower than the authorities had initially hoped. Surveys conducted at the time indicates that many were skeptical due to the risk of side effects already before the vaccination campaign. Another potential reason is that the swine flu spread slower than initially anticipated.

The public health authority estimates that the vaccination campaign saved around 100 lives and prevented 215 intensive care unit treatments.

The authorization of Pandemrix was fast-tracked due to exceptional circumstances, which allowed it to market with only data from the first phase of clinical trial in place. This was possible because of a mock-up vaccine route, where the complete vaccine protocol, including the adjuvant, had already been tested extensively, which only required the virus strain to be adjusted for development. Starting in May 2009 Glaxosmithkline (GSK) received orders from several countries to supply a vaccine against the swine flu. Sweden had already signed a public procurement contract with GSK in November 2007 in

place, stating that they would deliver influenza vaccine in the event of a new pandemic. The contract also prescribed that GSK would not be held responsible for potential side effects.

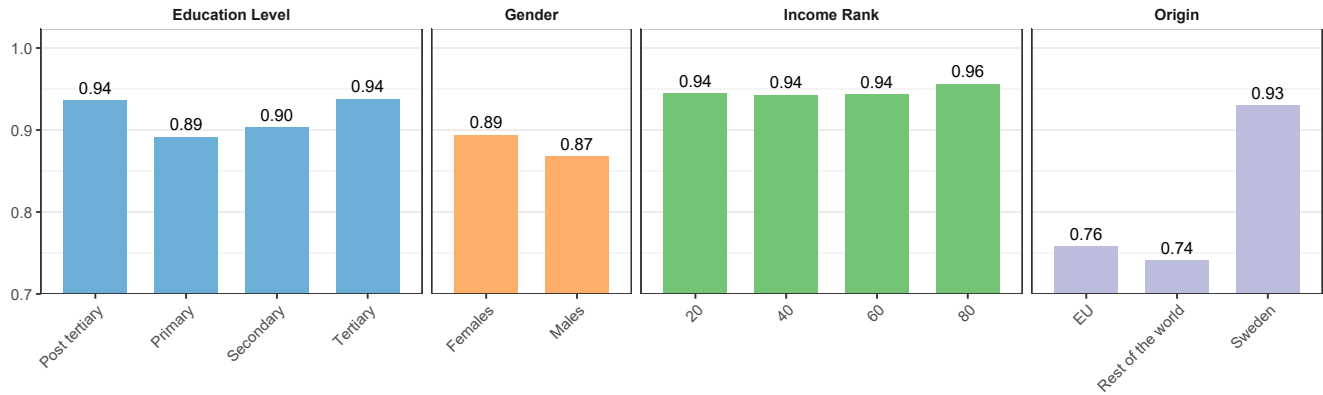
Appendix B. Additional Descriptive Statistics

Table B1: Predictions

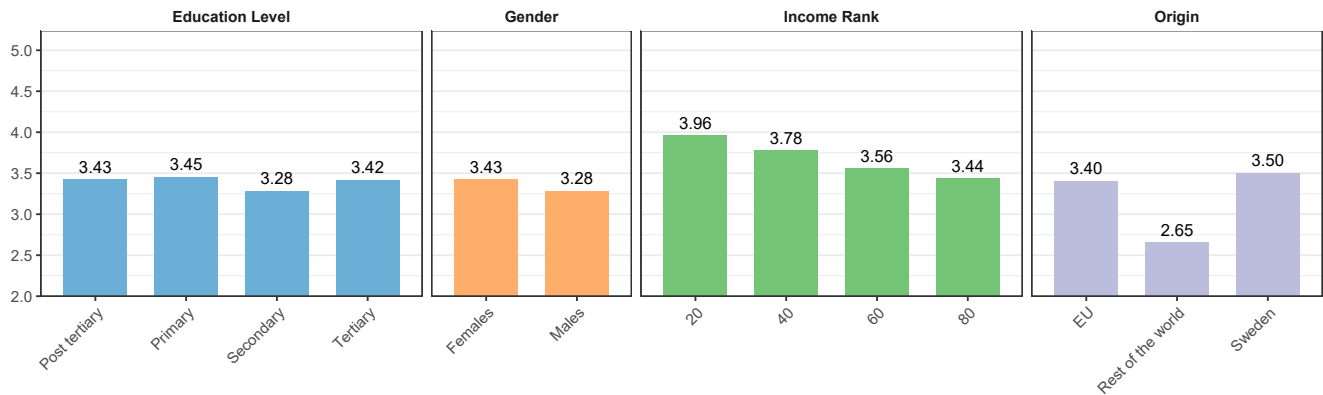
	Treated	Narcolepsy diagnosis	Pandemrix	Heart attack
Socioeconomic	0.52	0.55	0.53	0.58
Socioeconomic + Health	0.55	0.54	0.53	0.61

Notes: This table displays AUC-values for predicted probabilities. Probabilities are computed using XGBoost. First column compares treated units to individuals that received Pandemrix (control units). Second column compares individuals who are diagnosed with narcolepsy after 2016 and who hence were unlikely to develop it from Pandemrix to random individuals of the Swedish population. Third column compares individuals who take Pandemrix to individuals residing in the same regions. Fourth column compares individuals who are diagnosed with heart attack to random individuals of the Swedish population. The prediction exercises are run on samples that are balanced in birth year.

Vaccination Rate



Number of Doses



Days Elapsed

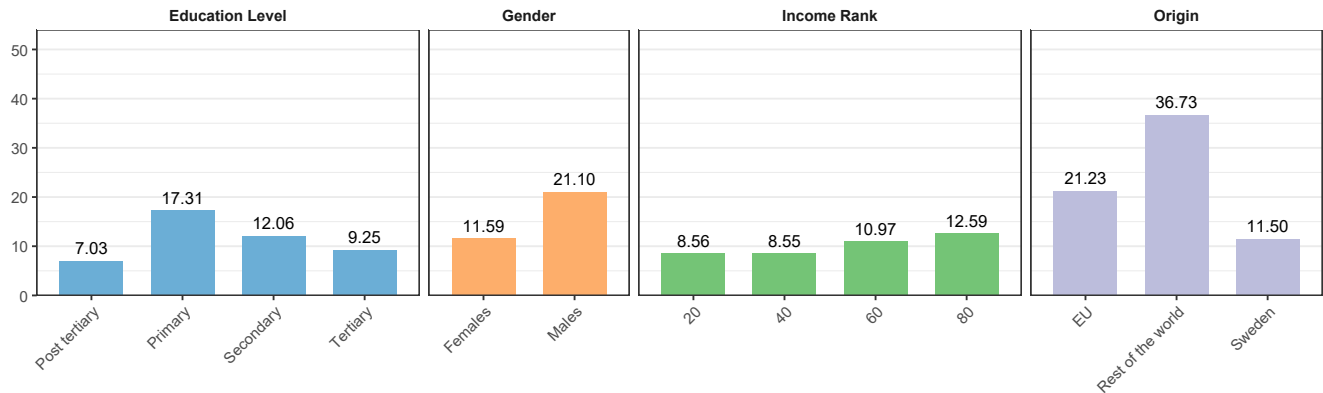


Figure B1: COVID-19 immunization outcomes for different sub-populations.

Notes: Each panel reports COVID-19 immunization outcomes for different sub-populations. **First column:** by education level (individuals aged 40–50 in 2022). **Second column:** by gender (individuals aged 40–50 in 2022). **Third column:** by income percentile (individuals aged 40–50 in 2022). **Fourth column:** by origin (Swedish vs. foreign background; foreign defined as being born abroad or having two parents born abroad; individuals born 2004 or earlier). Outcomes include vaccination rate (top row), average number of doses (middle row), and average days elapsed since first eligibility (bottom row).

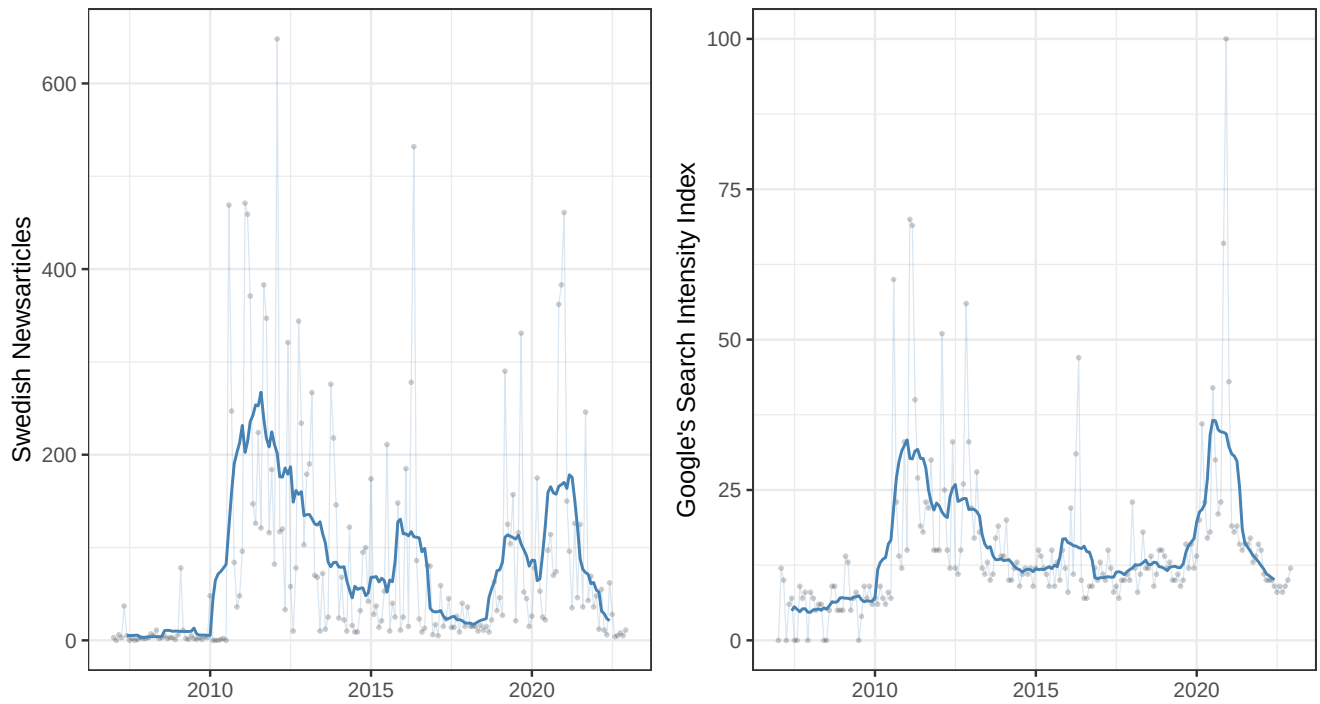


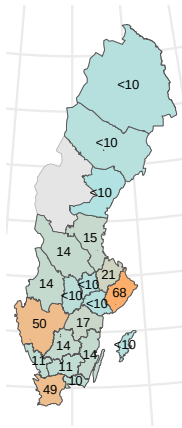
Figure B2: Exposure and Attention to the Narcolepsy Scandal

Notes: (a) Monthly number of articles mentioning "Narkolepsi" (eng. narcolepsy) in Swedish press. (b) Monthly Google search intensity for the word "Narkolepsi" (eng. narcolepsy). Each dot correspond to one month while thick line displays 12-month moving averages.

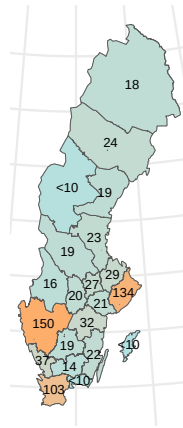


Figure B3: Narcolepsy Drug Prescriptions Across Time

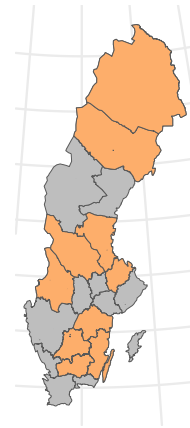
Notes: This figure shows the share of individuals prescribed each of twelve drugs commonly used to treat narcolepsy symptoms, by year and by sample group. The ATC subgroup N06 belongs to the anatomical group N for the nervous system and includes psychoanaleptics: N06A denotes antidepressants, while N06B denotes psychostimulants (including ADHD medications and nootropics). “Narcolepsy Post Swine flu” refers to individuals diagnosed with narcolepsy after the 2009–2010 swine flu pandemic, while “Narcolepsy Pre Swine flu” refers to those diagnosed earlier.



(a) post-swine flu Cases



(b) pre-swine flu Cases



(c) Pandemrix Vaccinations

Figure B4: Geographical Distribution of Individuals with Narcolepsy and Pandemrix.

Notes: (a) Number of treated individuals by healthcare region of residence in 2011. (b) Number of treated individuals by healthcare region of residence in 2011 in the pre-swine flu sample. (c) Healthcare regions with individual level Pandemrix data.

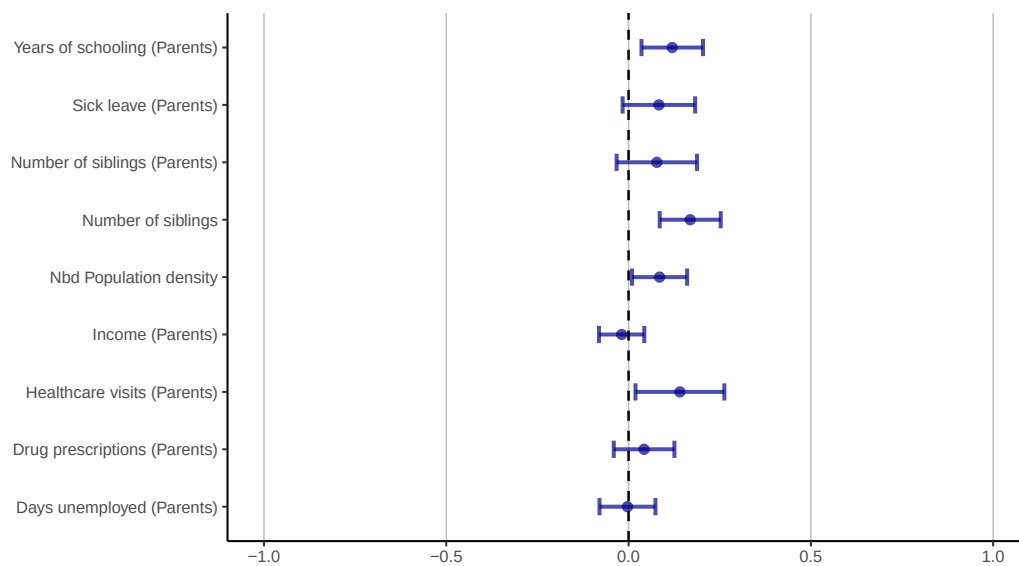
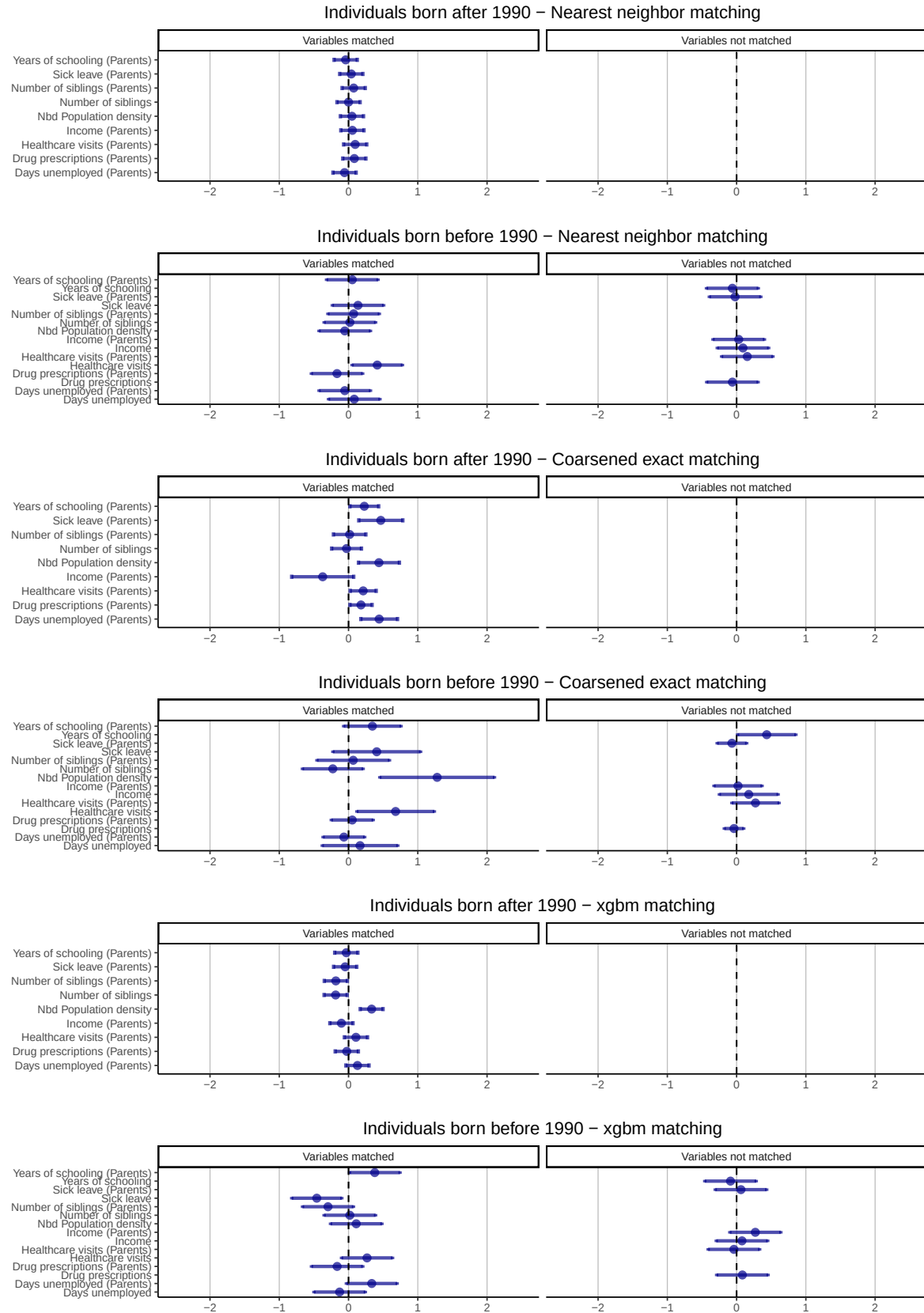


Figure B5: Balance in pre-treatment covariates, Placebo Control Sample

Notes: This figure displays differences in pre-determined characteristics between individuals that develop narcolepsy and control individuals for the pre sample. Each estimate comes from a univariate regression of the standardized covariate on the treatment status and on birth year fixed effects. Only a subset of covariates are displayed, the full pool of candidate covariates are listed in Appendix D.

Figure B6: Balance – Alternative Matching Methods



Notes: This figure displays differences in pre-determined characteristics between individuals that develop narcolepsy and control individuals for the main sample. Each estimate comes from a univariate regression of the standardized covariate on the treatment status. **Left subfigures:** Differences between treated and control units in the matched sample for matched variables. **Right panels:** Differences between treated and control units in the matched sample for variables not chosen for matching. Note that the variables used for matching is independent of the matching method.

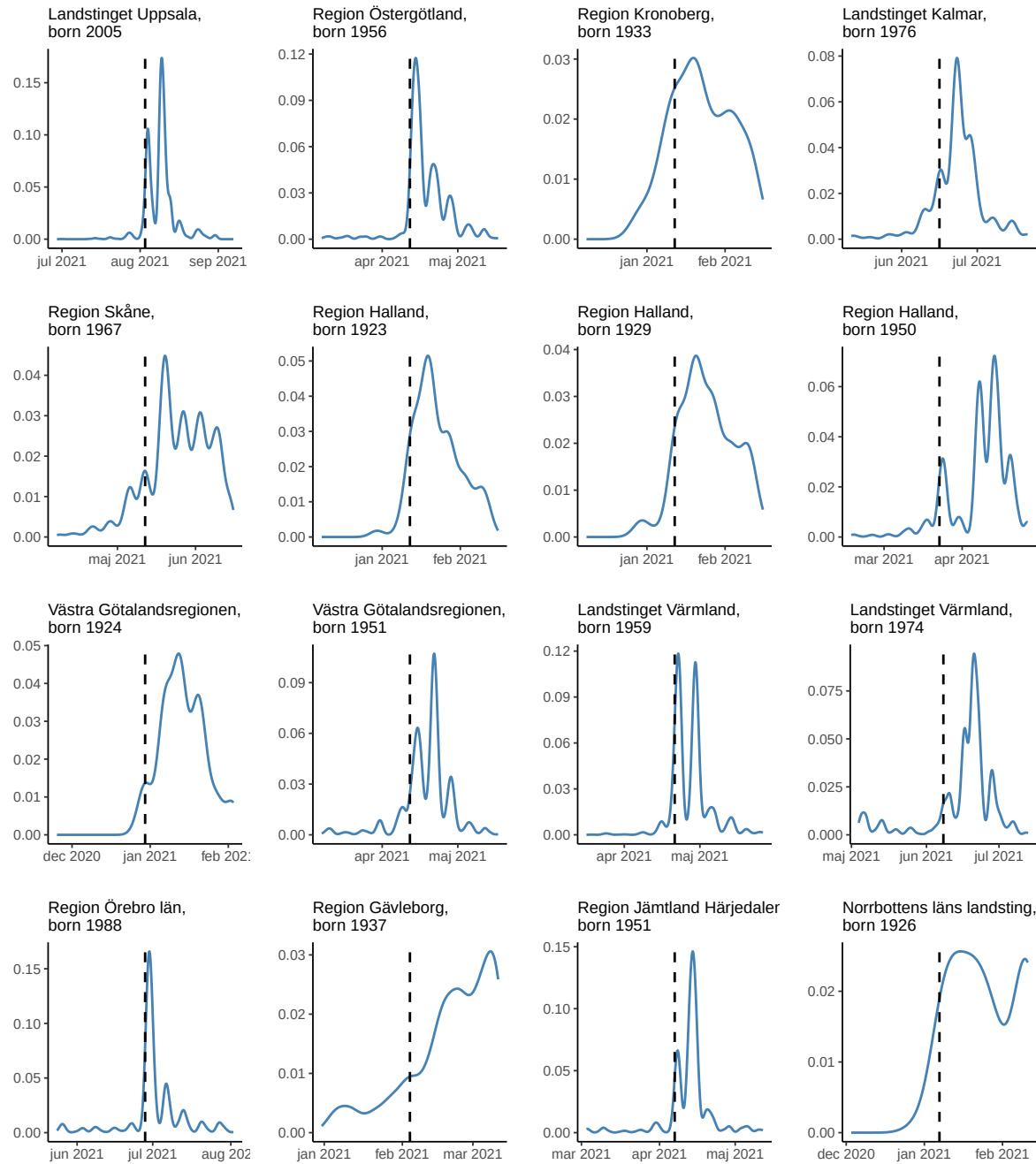


Figure B7: Definition of First Date of Availability

Notes: This figure displays the distribution of the first date an individual receives the COVID-19 vaccine for a sample of 16 Region×Birth-Year cells, showing a two-month window around the estimated date of vaccine availability (dashed line).

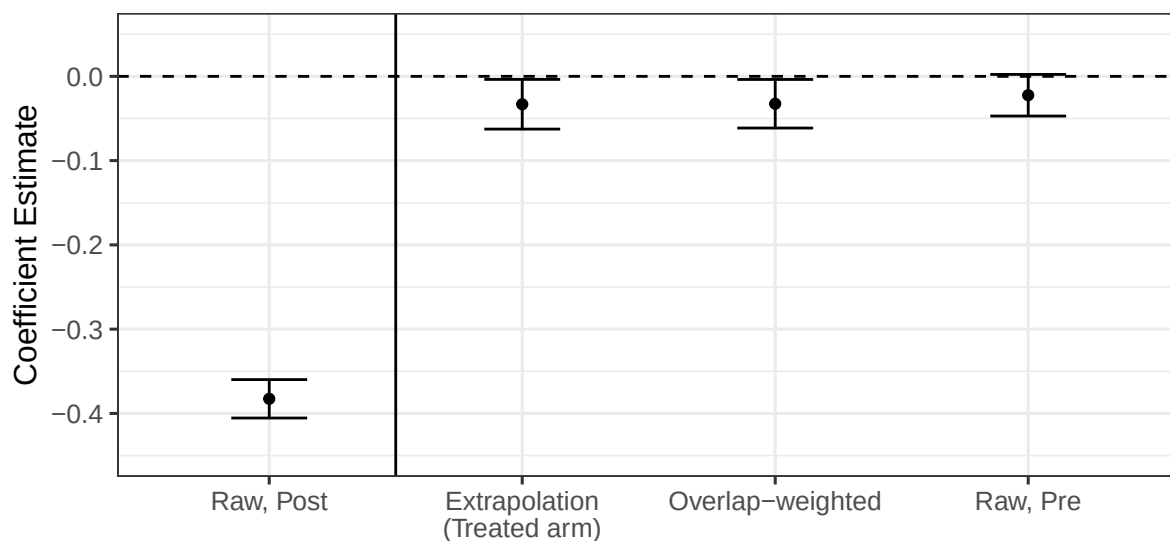
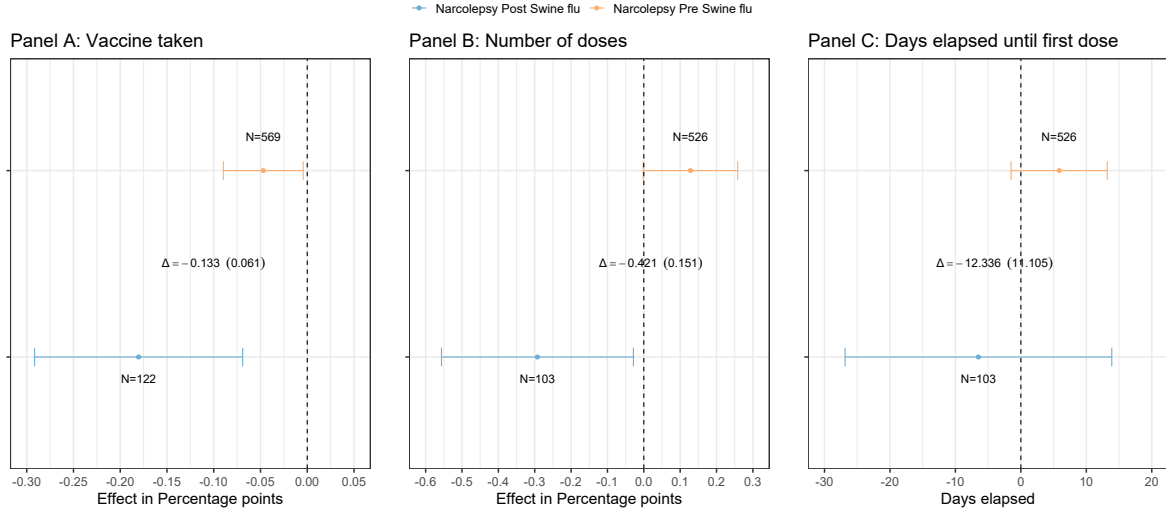


Figure B8: Counterfactual Narcolepsy Effect

Notes: **Raw, Post** is the unadjusted treated-control difference in the main sample. **Extrapolation (treated arm)** is the model-based treatment effect for target ages that lack treated-arm support in the placebo control group. We fit a natural spline in the placebo control group, predict $Y(1)$ and $Y(0)$ for the age distribution of individuals with vaccine induced narcolepsy. Confidence intervals use 100 bootstrap replications. **Overlap-weighted** reweights placebo control units to the age distribution of post swine flu individuals using density-ratio weights from $\Pr(S=1 \mid \text{age})$. **Raw, Pre** is the unadjusted treated-control difference in the placebo control sample.

Appendix C. Additional Results

Figure C1: Main results – Partners.

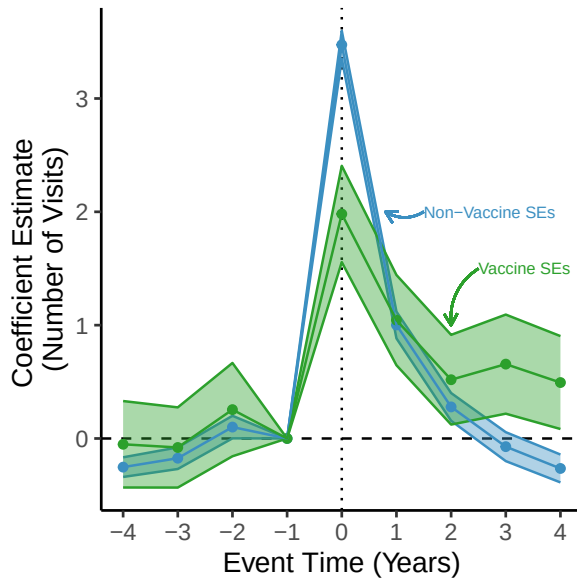


Notes: This figure displays coefficient corresponding to τ_{post} and τ_{pre} in eq. (4), similar to fig. 7 but for partners of the focal individuals. **Panel A:** Displays results for whether or not an individual has taken at least one COVID-19 dose. **Panel B:** Displays results for the number of COVID-19 doses, conditional on having taken at least one dose. **Panel C:** Displays results for the number of days elapsed until the first COVID-19 dose is taken. Standard errors are clustered by treatment cluster, i.e. by the treated or control individual that a network member is related to.

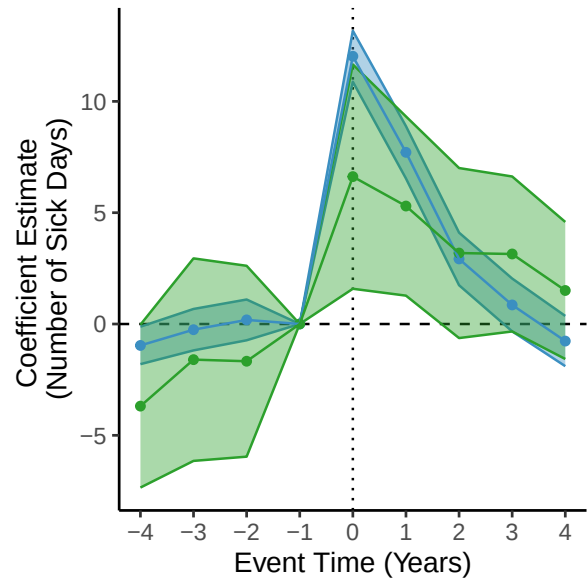
Table C1: Immunization outcomes and Mild Pandemrix Side Effects.

	Vaccine Taken	Number of Doses	Days Elapsed
Treated	-0.036 (0.008)	-0.061 (0.025)	-4.593 (1.847)
Num.Obs.	3907	3648	3648
R2	0.055	0.273	0.045

Notes: This table displays estimates from regressing each outcome variable on a binary indicator equal to 1 if an individual reported a mild side effect from pandemrix. Each treated individual is matched 1:1 to a control individual who also received Pandemrix but did not report any side effects. Matching is based on propensity scores estimated using pre-treatment health and socioeconomic characteristics. Matches are restricted to individuals of the same birth year and gender.



(a) Number of Healthcare Visits

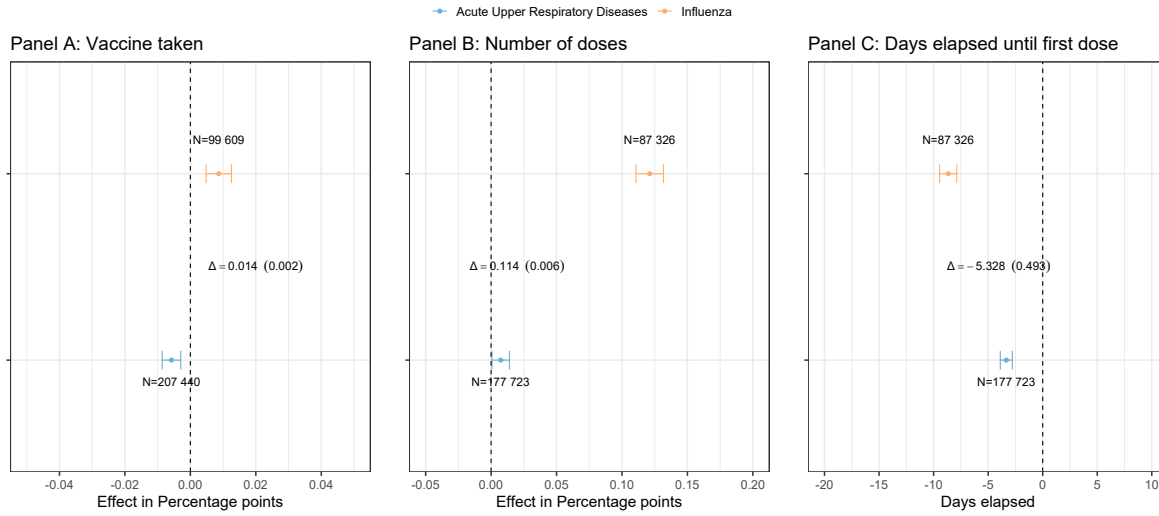


(b) Number of Sick Days

Figure C2: DiD-Estimates of Health Effects from Adverse Events

Notes: The figure shows estimated health outcomes around the reporting of a severe side effect relative to a matched control group for vaccine side effects and non-vaccine side effects. Panel (a) reports the number of visits to specialized healthcare; Panel (b) reports the number of sick days. Estimates are obtained from a standard TWFE specification.

Figure C3: Effect of Developing Influenza.



Notes: This figure compares individuals that received the main diagnosis i) Influenza ii) Upper respiratory tract infection in specialized healthcare at some point between 2015 and 2021 to matched individuals with similar characteristics of the Swedish population. The definition of main diagnosis requires the individual to be hospitalized due to the respective diagnosis and discards additional diagnoses. **Panel A:** Displays results for whether or not an individual has taken at least one COVID-19 dose. **Panel B:** Displays results for the number of COVID-19 doses, conditional on having taken at least one dose. **Panel C:** Displays results for the number of days elapsed until the first COVID-19 dose is taken. Standard errors are clustered by treatment cluster, i.e. by the treated or control individual that a network member is related to.

Table C2: Side Effects and Immunization Outcomes – Reweighted

	General Side Effects			Vaccine Side Effects		
	Vaccine Taken	Number of Doses	Days Elapsed	Vaccine Taken	Number of Doses	Days Elapsed
Diseased Individuals						
Side Effect	-0.019 (0.003)	-0.025 (0.013)	-1.90 (0.63)	-0.097 (0.013)	-0.062 (0.059)	11.2 (2.95)
N. Treated	15 294	13 974	13 974	849	686	686
Family Members						
Side Effect	-0.001 (0.002)	0.048 (0.009)	0.74 (0.38)	-0.042 (0.008)	-0.15 (0.037)	7.83 (1.54)
N. Treated	34 018	31 445	31 445	2 048	1 771	1 771
Children						
Side Effect	0.005 (0.003)	0.015 (0.008)	-0.77 (0.52)	-0.029 (0.018)	0.072 (0.044)	3.25 (2.89)
N. Treated	28 087	22 124	22 124	887	662	662
Partner						
Side Effect	-0.003 (0.003)	-0.015 (0.018)	-0.78 (0.72)	-0.036 (0.014)	-0.12 (0.086)	5.03 (4.15)
N. Treated	7 430	7 049	7 049	329	311	311

Notes: This table displays results from regressing the COVID-19 immunization outcome variables on a variable indicating if someone developed an reported a side effect between 2015 and 2020. Columns 1–3 use an indicator for reporting any side effect that was not vaccine-related between 2015 and 2020 as the regressor (“General side effects”). Columns 4–6 instead use an indicator for reporting a vaccine side effect (“Vaccine side effects”). Each treated individual is matched 1:1 to an untreated individual who (a) was born in the same year and (b) took the same drug in the same year as the treated individual. Weights are computed based on birth year using kernel smoothing.

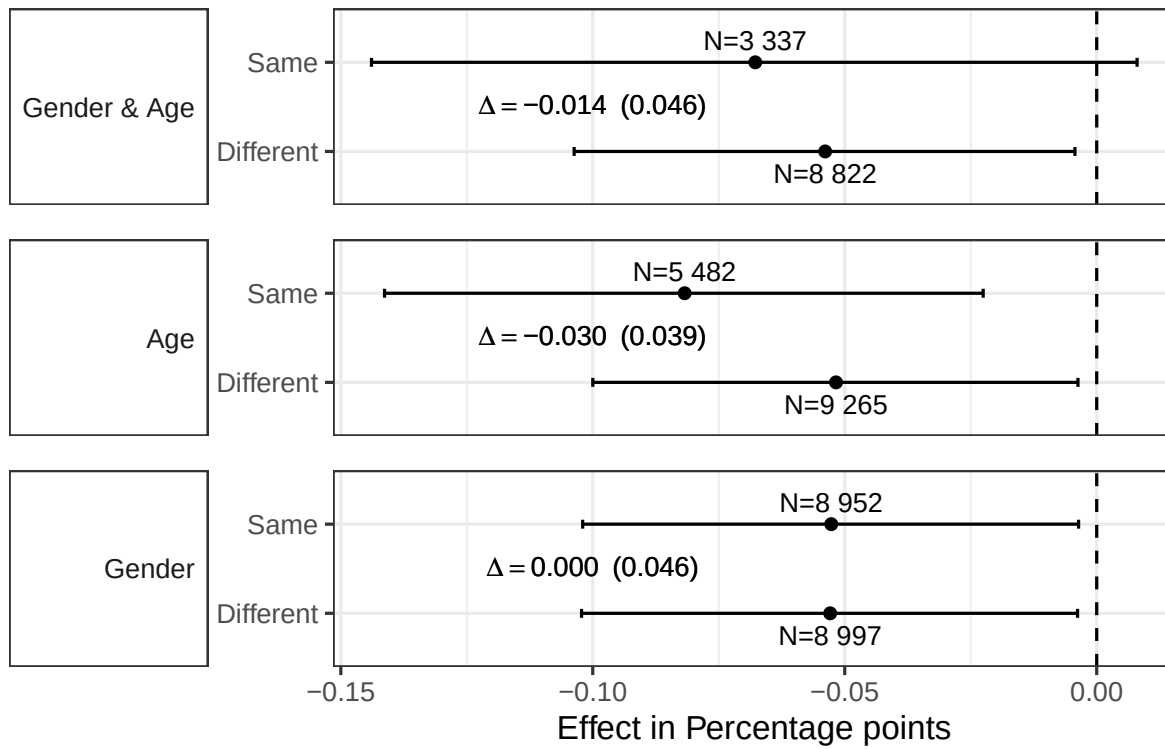
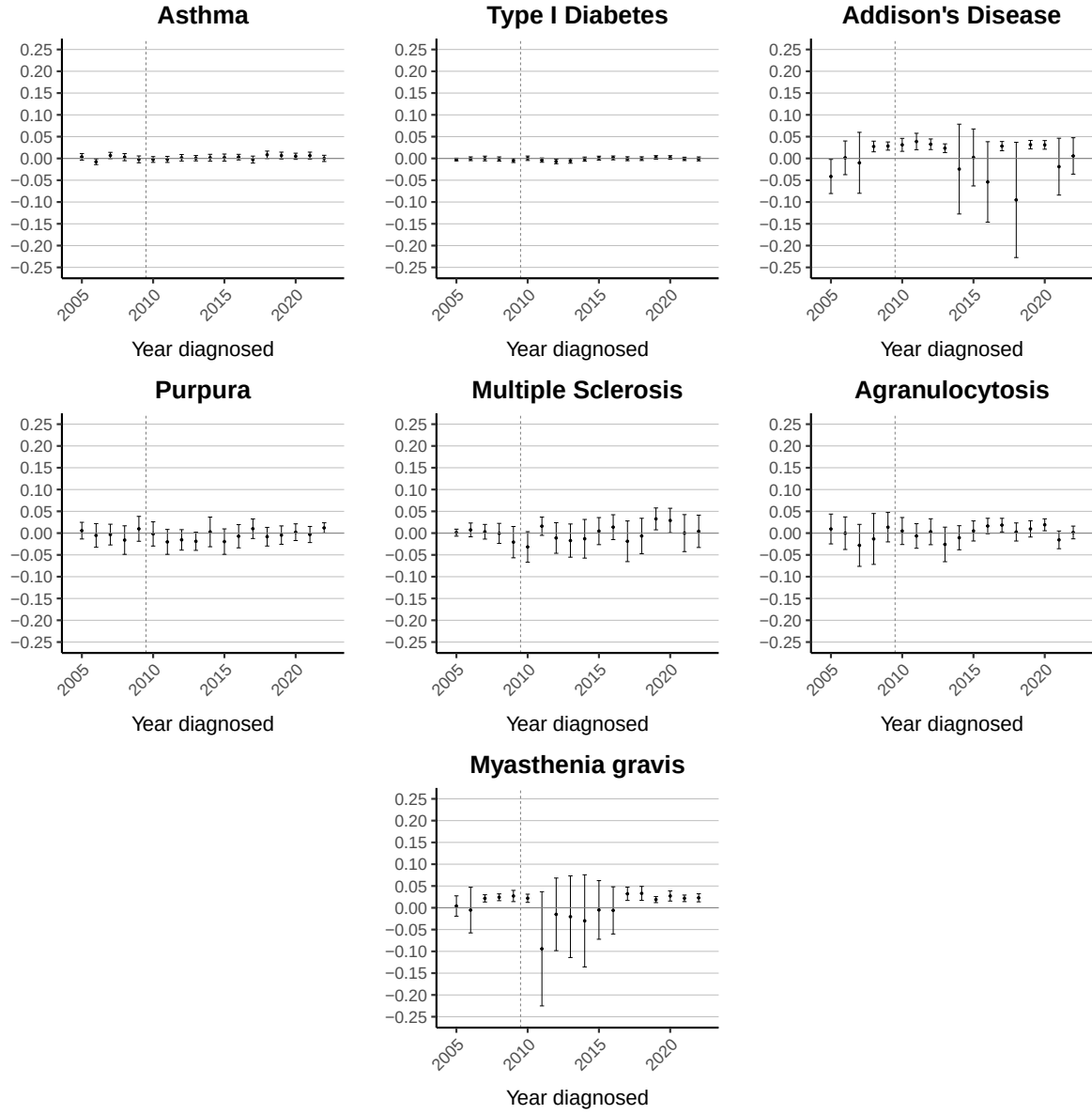


Figure C4: Heterogeneity – Social learning

Notes: This figure shows the coefficients for τ_{post} related to the variable *Vaccine Taken* among cousins of individuals in the focal sample. First row: Heterogeneity by gender. Second row: Heterogeneity by age, where individuals are considered the same age if their age difference is one year or less. Third row: Heterogeneity based on whether individuals share both gender and age.

Figure C5: Other Severe Diseases – Timeline



Notes: This figure shows the coefficients for τ_{post} for the variable *Vaccine Taken* for other diseases that people may have attributed to pandemrix. The control group consist of individuals who received pandemrix but did not developed narcolepsy. The treatment group consist of individuals who were first diagnosed with each of the respective diagnoses in year t . Each regression includes birth year fixed effects.

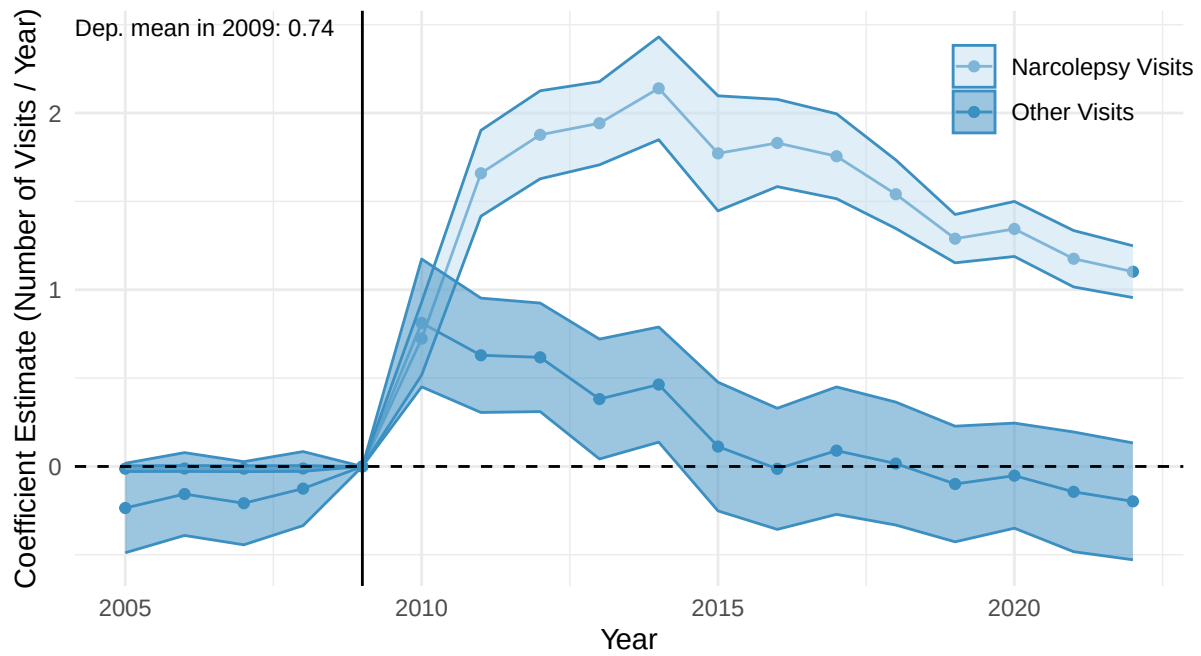


Figure C6: DiD-Estimates of Healthcare Visits After Narcolepsy Diagnosis

Notes: This figure displays difference-in-differences estimates from regressions where the outcome is the number of healthcare visits per year. The sample consists of treated individuals matched on birth year to others who also received pandemrix. Each treated individual is matched to three controls. The light blue line shows coefficients for visits where the main diagnosis is narcolepsy, while the dark blue line shows coefficients for visits related to other diagnoses. Standard errors are clustered at the individual level.

Appendix D. Appendix – Covariate definitions

We here define the set of candidate covariates used.

Table D1: Summary of Variables

Variable	Description
Birth year	Year of birth
Days sick	Yearly average number of days reported sick during 2005–2009
Gender	Gender of the individual
Income	Yearly average inflation adjusted disposable income during 2005–2009
Siblings	Number of full, biological siblings
Population density	Population density in individual's neighborhood in 2009
Origin	Indicator for either Sweden, Rest of Europe, or Rest of World depending on birth country of the individual and her parents
Number of drugs	Yearly average number of drugs prescribed during 2005–2009
Number of visits	Yearly average number of specialist healthcare visits during 2005–2009
Years of schooling	Highest level of schooling attained as of 2009 based on a mapping from the SUN classification (Svensk utbildningsnomenklatur) to number of years
Field of Education	Last field of education as of 2009 based on the SUN classification (Svensk utbildningsnomenklatur). 25 broad educational categories, defined based on degrees from high school, upper secondary school or university.
Drugs taken	14 binary variables. One for each level 1 ATC code. Equal to 1 if an individual received a drug within that ATC category at least once between 2005 and 2009.
Diagnoses	21 binary variables. One for each ICD-10 chapter. Equal to 1 if an individual was diagnosed within that Chapter category at least once between 2005 and 2009.

Appendix D. Appendix – Mild adverse events sample and regional Pandemrix data

Table D3 contains information about the reported mild symptoms. 63% of reported symptoms from Pandemrix are classified as mild, primarily consisting of symptoms like febrile disorders, connective tissue pain, headache, or others. 54% of side effects are self reported, the remaining reports are made by doctors and nurses. In our preferred specification we include both individuals reported side effects themselves and individuals that had a doctor or a nurse report the side effect for them.

Table D2: Reported mild side effects.

	Symptom	Share of reports	Self reported	Professionally reported
Musculoskeletal and connective tissue pain and discomfort	General signs and symptoms NEC	0.10	0.70	0.30
	Febrile disorders	0.06	0.53	0.47
	Injection site reactions	0.06	1.00	0.00
	Pain and discomfort NEC	0.05	0.44	0.56
	Headaches NEC	0.05	0.66	0.34
	Asthenic conditions	0.04	0.71	0.29
	Nausea and vomiting symptoms	0.04	0.53	0.47
	Neurological signs and symptoms NEC	0.04	0.60	0.40
	Paraesthesias and dysaesthesias	0.04	0.55	0.45
	Urticarias	0.03	0.16	0.84
	Muscle pains	0.02	0.54	0.46
	Joint related signs and symptoms	0.02	0.68	0.32
	Feelings and sensations NEC	0.02	0.75	0.25
	Rashes, eruptions and exanthems NEC	0.02	0.26	0.74
	Others	0.33	0.42	0.58

Notes: This table displays the most common reported non-severe side effects by the 2796 individuals who only reported mild symptoms along with the share of each symptom that was self reported and reported by professionals for the same individuals. We remove instances where an individual report the same symptom multiple times.

Table D3: Reported adverse events

All Side Effects		Vaccine Side Effects	
Symptom	N. Reports	Symptom	N. Reports
Central nervous system haemorrhages and cerebrovascular accidents	434	Vaccination site reactions	74
Therapeutic and nontherapeutic responses	370	Disturbances in consciousness NEC	32
Angioedemas	327	Febrile disorders	28
Nausea and vomiting symptoms	284	Injection site reactions	26
Urticarias	276	Musculoskeletal and connective tissue pain and discomfort	24
Disturbances in consciousness NEC	264	Paraesthesias and dysaesthesias	23
Anaphylactic and anaphylactoid responses	262	Nausea and vomiting symptoms	22
Breathing abnormalities	244	General signs and symptoms NEC	21
Rashes, eruptions and exanthems NEC	225	Headaches NEC	21
Neurological signs and symptoms NEC	219	Joint related signs and symptoms	21
Asthenic conditions	197	Seizures and seizure disorders NEC	21
Renal failure and impairment	171	Rashes, eruptions and exanthems NEC	19
Allergic conditions NEC	170	Neurological signs and symptoms NEC	17
Diarrhoea (excl infective)	162	Muscle pains	15
Hepatobiliary function diagnostic procedures	154	Urticarias	15
Others	11 362	Others	421

Notes: This table displays the most common reported adverse events between 01 January 2014 and 31 December 2020 excluding events related to influenza vaccines (ATC J07BB02). Column 1 represents non-vaccine related adverse events. Column 2 restricts to vaccine-related events. We remove instances where an individual report the same symptom multiple times.

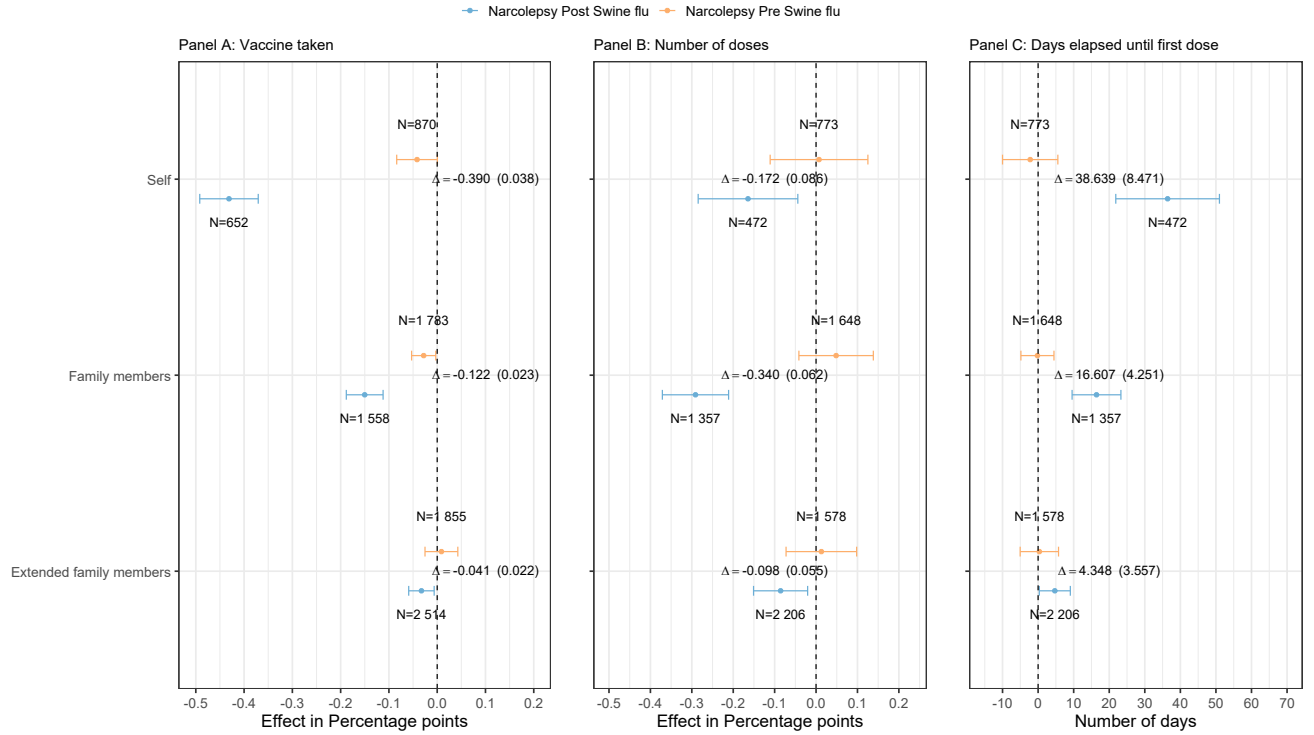
Table D4: Regional Pandemrix data

Region	N obs	Vaccination rate	Comment
Dalarna	167,855	0.61	
Gävleborg	175,239	0.63	
Jönköping	231,374	0.69	
Kalmar	78,113	0.33	Only vaccinations given in primary care facilities.
Kronoberg	66,488	0.36	Only vaccinations given in primary care facilities.
Norrbottn	131,195	0.53	
Uppsala	181,461	0.54	
Värmland	116,931	0.43	
Västerbotten	3,525	0.01	Only vaccinations from 2010-10-01 and onwards.
Östergötland	274,405	0.64	

Notes: Vaccination rates are calculated as the share of individuals residing in each healthcare region in 2009.

Appendix E. Robustness of Main results

Figure E1: Main results – Diseased individuals and family members, XGBM Matching.



Notes: This figure displays coefficient corresponding to τ_{post} and τ_{pre} in eq. (4). First row corresponds to coefficient for individuals that themselves developed narcolepsy (Self). The second and third rows show the corresponding coefficients for Family members and Extended family members respectively. **Panel A:** Displays results for whether or not an individual has taken at least one COVID-19 dose. **Panel B:** Displays results for the number of COVID-19 doses, conditional on having taken at least one dose. **Panel C:** Displays results for the number of days elapsed until the first COVID-19 dose is taken. Standard errors are clustered by treatment cluster, i.e. by the treated or control individual that a network member is related to.

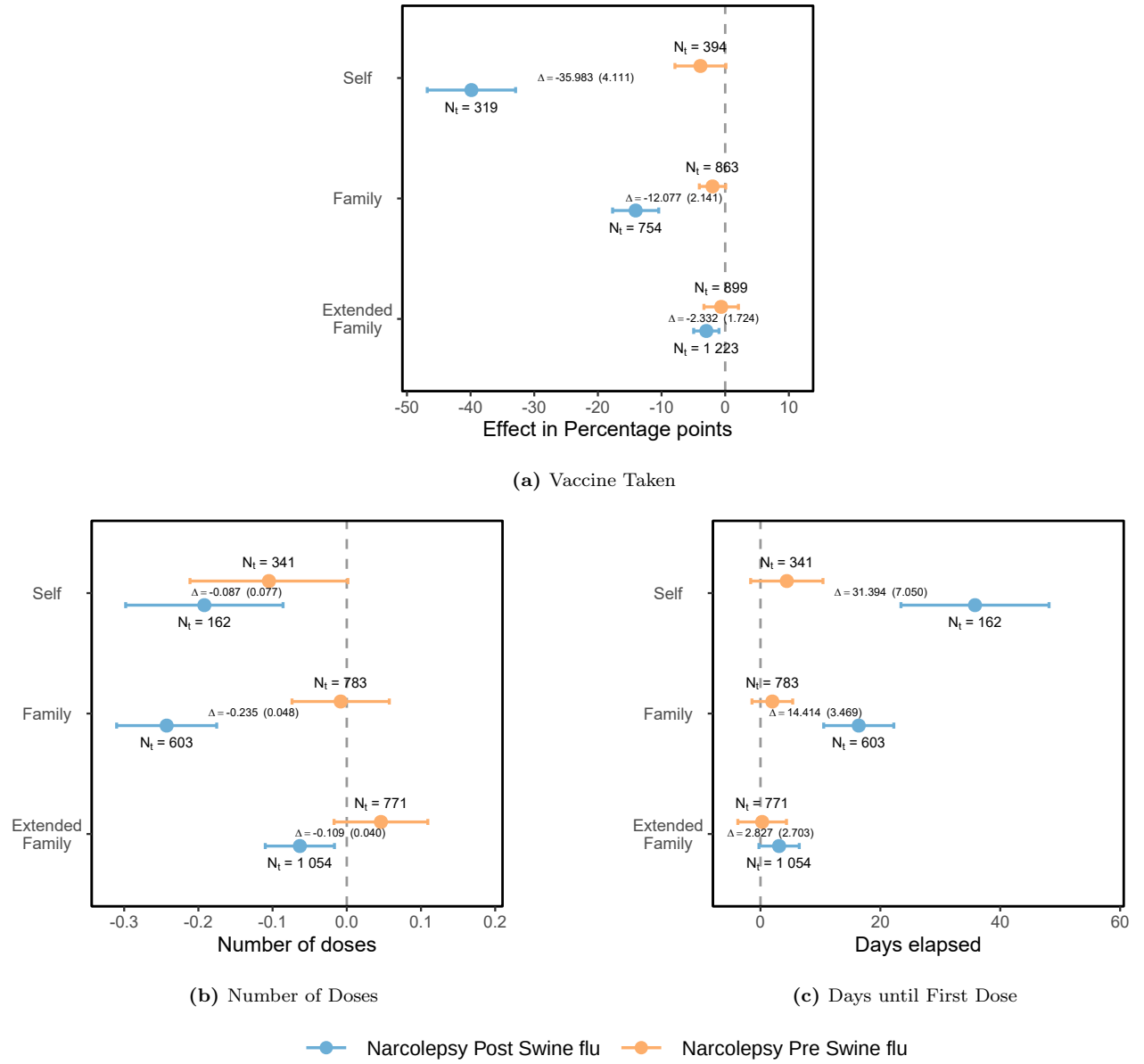


Figure E2: Main Results – Diseased Individuals and Family Members, CEM

Notes: This figure displays coefficients corresponding to τ_{post} and τ_{pre} in eq. (4). First row corresponds to individuals that themselves developed narcolepsy (Self). The second and third rows show the corresponding coefficients for Family members, defined as siblings and parents, and Extended family members, defined as aunts/uncles and cousins. **Panel A:** Ever taken at least one COVID-19 dose. **Panel B:** Number of doses, conditional on at least one dose. **Panel C:** Days elapsed until the first COVID-19 dose. Standard errors are clustered by treatment cluster, i.e. by the treated or control individual that a network member is related to.

Appendix F. Description of matching and predictions

F.1 Predictions in Table B1

For each prediction, each treated unit is matched to 100 control units with the same birth year. Control units consist of the Swedish population as of 2021.

Table F1: XGBoost hyper-parameters, cross-validation setup, and features

Model hyper-parameters	
Maximum tree depth (<code>max_depth</code>)	4
Learning rate (η)	0.10
Minimum child weight (<code>min_child_weight</code>)	1
Subsample fraction used for training (<code>subsample</code>)	0.8
Number of boosting rounds (<code>n_rounds</code>)	100
Model setup	
Cross-validation folds	5
Number of treated units	1 013
Features	
Socioeconomic: income, years of schooling, field of education, days unemployed, parental income, days on sick leave, parental years of schooling, number of siblings, parental number of siblings, parental days on sick leave, gender, world region of origin.	
Health: 4-digit level drug codes (ATC), 3-digit level diagnosis codes (ICD 10-SE).	

F.2 Predicting who takes the COVID-19 vaccine

We predict COVID-19 vaccination using a hybrid model that combines a sequence encoder for diagnosis histories with gradient-boosted trees on time-invariant covariates. Each individual’s diagnosis history is converted into an integer sequence, embedded and passed through a bidirectional LSTM, and projected into a low-dimensional representation. The encoder is trained with binary cross-entropy loss and a sigmoid output, using a 10% hold-out split for validation (batch size 256, 5 epochs). The resulting sequence embedding is concatenated with imputed and encoded socioeconomic covariates and fed into an XGBoost classifier trained with logistic loss. Final performance is evaluated on the 10% hold-out test set.

Appendix H. Mathematical Derivations

since $D = 1 \iff p_i > \bar{p}$, $\text{mar}(1)$ is equivalent to

$$D \perp Y(1) \mid W = 1, X.$$

which yields that

$$E[Y(1) \mid W = 1, D = d, X] = E[Y(1) \mid W = 1, X] \quad (d = 0, 1).$$

The same argument for $\text{mar}(2)$ implies that

$$E[Y(0) \mid W = 0, D = d, X] = E[Y(0) \mid W = 0, X] \quad (d = 0, 1).$$

MAR(1) and reported side-effect equivalence directly give us that

$$E[Y(1) \mid D = 1, X] = E[Y(1) \mid W = 1, D = 1, X] = E[Y(1) \mid W = 1, X]$$

What about $E[Y(0) \mid D = 0, X]$? We can write:

$$\begin{aligned} E[Y \mid D = 0, X] &= \Pr(W = 1 \mid D = 0, X) E[Y(1) \mid W = 1, D = 0, X] \\ &\quad + \Pr(W = 0 \mid D = 0, X) E[Y(0) \mid W = 0, D = 0, X]. \end{aligned}$$

In words, it is a weighted average of false negatives and true negatives. Bayes formula yields that.

$$\Pr(W = 1 \mid D = 0, X) = \frac{\Pr(D = 0 \mid W = 1, X) \Pr(W = 1 \mid X)}{\Pr(D = 0 \mid X)}.$$

We maintain the assumption that $\Pr(W = 1 \mid X)$ is small relative to the number of non-reporters such that the term vanishes. Together with $\text{mar}(2)$ it directly follows that

$$E[Y \mid D = 0, X] \approx E[Y \mid D = 0, W = 0, X] = E[Y \mid W = 0, X]$$

Table F2: RNN + XGBoost hyper-parameters, data splits, and features

Sequence encoder (RNN)	
Input	Label-encoded diagnosis codes (with mask_zero)
Vocabulary size	998
Sequence length	$\max\{50, 99\text{th percentile of lengths}\}$; post padding/truncation
Embedding layer	Dimension = 32
Recurrent layer	Bidirectional LSTM, 64 units per direction
Dense projection	32 units, ReLU
Output head	1 unit, sigmoid
Loss / optimizer	Binary cross-entropy / Adam
Batch size / epochs	256 / 5
Validation split	10% hold-out (stratified)
Gradient boosting classifier (XGBoost)	
Objective	<code>binary:logistic</code> (logistic loss)
Boosting rounds (<code>n_estimators</code>)	4000 (with early stopping)
Learning rate (η)	0.02
Max depth	5
Subsample / <code>colsample_bytree</code>	0.8 / 0.8
Regularization	$\lambda = 1.0, \alpha = 0.0$
Class imbalance	<code>scale_pos_weight</code> = $n_{\text{neg}}/n_{\text{pos}}$
Early stopping	200 rounds on a 10% validation slice of training data
Model setup	
Train / test split	90% / 10% (stratified)
Hybrid features	Concatenate sequence embedding (32-D) with tabular covariates
Feature sets	
Socioeconomic	income, years of schooling, field of education, days unemployed, parental income, days on sick leave, parental years of schooling, number of siblings, parental number of siblings, parental days on sick leave, gender, world region of origin
Health	4-digit ATC drug codes (sequence input), 3-digit ICD-10-SE diagnoses (sequence input), reported side effects (sequence input)